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City of Malolos, Bulacan

Library Management System Using RFID and Barcode

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ABSTRACT

The Library Management System Using RFID and Barcode is a system that iterates on the common library system in issuing and returning books. This system adds a way to log anyone in and out of the library using an RFID tag. The same tag is used to by the user to checkout books out of the library. The RFID tags are read by a RFID Reader attached to an Arduino Uno. The books are tracked using a barcode attached to the books. This would be read by a barcode scanner. The system also includes a database using Microsoft Access to contain all this data. The system is packaged on to a single program built using Visual Basic .NET. This system would simplify and make the process of borrowing and returning books more efficient. The results the researchers have gotten show that the system is viable and effective.



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**COMPUTER COLLEGE OF MALOLOS
COLLEGE OF ENGINEERING**

DEDICATION

This humble work is dedicated to our families, friends, classmates, future researchers and to all we disturbed during evaluation.



APPROVAL LETTER

This research proposal entitled “Library Management System Using RFID and Barcode”, proposed and submitted by Melvin C. Dela Cruz, Carlo Dominic M. De Guzman, John Philip N. Tuazon and Joash Christian B. Liwanag in partial fulfilment of the course requirements for the degree of Bachelor of Science in Electronics Engineering, has been examined and recommended for acceptance and approval for Oral Examination.

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

A library plays a very important role in promoting progress of knowledge. The library occupies a very important place in school. With its books suited to the interest and aptitude of students of different age group, with its magazines, periodicals, newspapers and with its calm and tranquil atmosphere it has a special call to the students who went there to for reading the materials which cannot be provided in classroom. Here they found an environment which is conducive to the building up of habit of self-learning.

Using manual management in the library slows down the process of logging in and out of students and the borrowing and returning of books. The lack of Information of Reports and Updates with respect to library transactions and services are not well disseminated. The issuing of book is slow where borrowers (faculty, students, staff and clients) will have to wait for further library transactions. Slow retrieval of Data viewing of library reports and files in terms of issuing and returning of books were not well-established/implemented. The library services in terms of issuing of books through library cards consumed a lot of time. Manual management make things complicated and inconvenient.

That is the reason why the researchers proposed a system that reduced the processing time of borrowing and returning of books and the logging of the students with



the use of barcode and radio frequency identification technology for academic libraries. This was done by the use of an Arduino microprocessor with a RFID reader to read and identify the RFID tags attached the user's library card and a barcode reader to read and identify books being borrowed. From the Arduino the data is to be sent using a universal serial bus (USB) to the server's computer and to be processed in the graphical user interface (GUI) that will be created using Visual Studio. The processed data will be displayed into a database which is to be created using Microsoft access. The books and the students can be registered into the data base, the users can also see a catalogue of books in the database for them to search for the book/s that they were looking and the time of the books that were borrowed will be logged into the database and if the book was returned but either damaged or overdue it will cost a penalty or fine. The time limit and the fine can be set by the researchers on the installation of the said device.

1.2 PROBLEM STATEMENT

Manual management consumes so much time and effort leading to worthless exhaustion and human error that usually creates problems in managing the library and is also slows down the process such as logging in and out of students, borrowing and returning of the books. A Library Management System using Barcode and RFID is needed, to reduce the processing time and to make things more convenient for the students and the librarian.

1.3 OBJECTIVE OF THE STUDY

1.3.1 GENERAL OBJECTIVES

- To create a library management system that will help in managing an academic library using Radio frequency identification (RFID) and barcode technology. It helps to automate borrowing and returning processes and allows identification of students and tagged objects like books, using radio waves and barcode.

1.3.2 SPECIFIC OBJECTIVES

1. To be able to create a database system using Visual Basic.
 - 1.1 To create Software based catalogue
 - 1.1.1 To help the student to search for specific book they need if available or not.
 - 1.1.2 To create a book inventory using database.
 - 1.2 To create a student Attendance logging system.
2. To be able to create a device using Radio Frequency Identification (RFID) and barcode technology.
 - 2.1 To directly provides the book information and library member information to the Library management System.
 - 2.2 To borrow books using barcode and RFID technology.
 - 2.3 To return books using barcode only.
 - 2.4 To log the users who enter the library easily using Radio Frequency Identification (RFID).
3. To be able to create a device using Arduino Uno.
 - 3.1 To send data to process the server through a Local Area Network.
 - 3.2 To receive data from the Radio Frequency Identification (RFID) and barcode.



4. To be able to create a GUI (graphical user interface) using Visual Basic to be the Platform of the database.

4.1 To be able to create database using Microsoft access

1.4 SCOPE AND DELIMITATION

1.4.1 SCOPE

The study covers the use of the library by making it more convenient for both the user and librarians by streamlining the process of borrowing a book from the library. This is done through the use of barcode tags.

The study requires both the user and the books to be registered to a central database. This means that a new user is provided a way to register into the system to access the library.

The study provides a searchable database running on Microsoft access which contains all the information regarding the books and who borrowed them. This database is searchable by anyone who is inside the library.

The study also provides a graphical user interface (GUI) to be the platform of the database to process the data from the arduino using visual basic.

Each book supposed to have a barcode tag. To borrow a book, the user must bring the book and be scanned at a barcode reader connected to an Arduino. The Arduino in turn sends data to process the server through a Local Area Network.

The study also provided a way on monitoring who goes in and out of the library with the use of a library card that contains information about the user. This library card is



also equipped with a RFID tag. The users have to be registered to have (RFID) to use for logging and borrowing/returning of books.

1.4.2 DELIMITATION

There are some limitations implementing the device. The active Radio Frequency Identification (RFID) tag has limitations on the lifetime of the tag (5 years), so the consumer needs to change the tag every time the tag expires, the researchers may conduct a maintenance for the system and to change the RFID tags. RFID tag does not contain a power source; the power is supplied by the reader. The passive tags can be read only at very short distances, typically a few feet at most. The barcode tags can only be read in a few centimetres. The location of the book cannot be tracked because the researchers did not make use of RFID Tracking tags. This device is not concerned about lost and misplaced items of the library and cannot notice if a student has released the book in the library without scanning the barcode and RFID. This device is not supported by internet, so they are only connected through Local Area Network.

1.5 SIGNIFICANCE OF THE STUDY

This system is designed to make library's processes easier and more convenient to the users and also to the staffs of the library, considering the hassle of queuing in library just to login and the time consuming of searching books. The following is the benefits of this system.



To the Students, The students can easily login by just tapping their library card and they can easily borrow books. And it can also save time for the students, who are searching for the book they need.

To the Librarian, The system makes the librarian's job easier. They no longer need to issue forms for the students and make soft copies of the forms to borrow/return books, to register, and to log in and log out to the library.

To the Schools, The schools make students comfortable in using library. And it is also advantage for them as they offer modern way of using library.

To the Researchers, This system can be serve as their reference or guide in making their own research that can help solve problems in our society.

To the Future Researchers, this system will help them as guide in their system that will solve future problem in the future society.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 RELATED LITERATURE

2.1.1 FOREIGN LITERATURE

Radio-frequency identification devices are a daily part of the electronic age — found in passports, and library and payment cards. Eventually they're expected to replace barcode labels on consumer goods. Now schools across. Like most state-financed schools, the district's budget is tied to average daily attendance. If a student is not in his seat during morning roll call, the district doesn't receive daily funding for that pupil because the school has no way of knowing for sure if the student is there. But with the RFID tracking, students not at their desk but tracked on campus are counted as being in school that day, and the district receives its daily allotment for that student.

Each active-tag reader captures the ID number of every badge within range and transmits that information to the Wade Garcia software. The badge's active tag is typically set to beacon its ID number every 28 seconds. In the system's software, a tag's ID is linked to the name of the individual assigned to that badge, as well as to some details, such as whether he or she is a student or staff member—and, if that person is a



student, his or her grade level. The district can log into the software at any time and views the locations of specific individuals on the premises.

The idea is to install card readers in all classrooms that seat 50 or more students -- those in which it'd be difficult or time-consuming to take attendance every day. Professors who teach smaller classes would have to find another way to take attendance. Given that class sizes shrink as a student's education progresses, the policy would likely impact freshmen and sophomores. Students wouldn't have to take their ID cards out, because the readers would be sensitive enough to pick up signals from anywhere in the room. Data from each class would be recorded, and instructors would get reports on who was in class, the Arizona Republic reports.

As part of a controversial trial that could someday include 112 schools and nearly 100,000 students, North side Independent School District in San Antonio, Texas, recently issued students at two of its campuses new badges with an embedded RFID (radio-frequency identification) chip in order to track their locations. Unlike passive chips that transmit data only when scanned by a reader, these chips have batteries and broadcast a constant signal so they can track students' exact locations on school property, down to where they're sitting—whether it's at a desk, in a counsellor's office, or on the toilet.

Grade-school students in a north-eastern Brazilian city are using uniforms embedded with locator chips that help alert parents if they're cutting classes, the city education secretary said Thursday. Twenty thousand students in 25 of Vitoria da Conquistas 213 public schools started using T-shirts with chips earlier this week, Secretary Coriolano Moraess aid. By 2013, all of the city's 43,000 public school students,



aged 4 to 14, will be using the chip-embedded T-shirts, he added. Radio frequency chips in "intelligent uniforms" let a computer know when children enter school and it sends a text message to their cell phones. Parents are also alerted if kids don't show up 20 minutes after classes begin with the following message: "Your child has still not arrived at school".

According to Robson (2001) usability is a key requirement for users, says Elisabeth Robson, Product manager for Online Computer Library Center. The catalogue has become a way to pull together disparate resources, including commercial resources and web links. Management systems also allow circulation, including check in/check outland enable libraries to purchase materials and track where they are.

According to Primer (2005) for many businesses automating the collecting of employee attendance data when employees begin their shifts is still a manual process. Even business that has automated or computerized processes such as scheduling, inventory, purchasing, general ledger and payroll processing still use manual methods such as timecards or attendance sheets to collect time and attendance data.

According to E. Valentin (2006) from the article entitled D. "The State of Library Management Systems" are systems that helps big in terms of some transactions on the library. On this article the automated library systems said that it can do multitasking. It can do recording books on library, but it can also make direct transactions It will make the library service convenient like making card cataloging. Therefore, multitasking inside the library is very nice. He system will be the one who records all the transactions inside

the library so it decreases the tasks of the librarian so it can avoid the cases of lost records due to human errors.

Simmond (2001) highlights the role of electronic information resources on collection management in academic libraries. He says that the advent of electronic information resources created new methods of document delivery and access to information. With the application of internet and the use of new information technologies the range of services provided by academic libraries increased dramatically. He also stressed that users can access the libraries resources without stepping into library building. Further they can also access other library resources like online catalogues and freely available databases.

Based on Henderson (2008), Inventory System Consist: An inventory control system is used in many warehouses to track location, quantity and status of goods that are to be sold. Shipping and receiving of products are scheduled by and inventory control system. Inventory control uses barcodes and RFID tags to track specific items. An audit, or a physical count of inventory, is used to find out how many items are in an inventory. Inventory taking is often required in order to get an accurate count for accounting valuation purposes. Some inventory items have barcodes attached to them or imprinted on them. These barcodes can be read during audit by barcode reading devices.

2.1.2 LOCAL LITERATURE

A librarian demonstrates how to use the Self-Service Library Check-out Station, where students can check out books themselves. Among the library processes that will benefit from the UHF RFID solution are the circulation services, inventory, and security



system. The solution can provide real-time statistics of both Intramuros and Makati libraries' patron visits and can generate inventory reports instantly. "This UHF-RFID system is very efficient and a logical move for us because we are a technological school," said Lansangan. "Apart from making our processes here efficient, this is also to keep up with the technology employed in the US.

The Citra Metro Manila Toll Ways Corporation has begun its phasing out of the old e-Pass and will be replacing it with RFID passes. CMMTC has posted guidelines on how to get the new e-Card system with the first 200,000 applicants getting them for free. The RFID consists of the e-Card itself as well as two kinds of RFID stickers to be placed on the car. The first option will be to place the sticker on the corner of the headlight or where it doesn't obstruct the beam of light. The other will be placed in the windshield of the car. According to a report by Inquirer, the system will be able to scan a vehicle even at a speed of 50 kilometers per hour.

"We believe that the RFID technology can complement the DOST's Smarter Philippines program, the essence of which is the effective generation, gathering, and analysis of data to enable timely and effective decision making and planning," said Louis Casambre, Executive Director, ICT-Office. At present, RFIDs are being used in DOST's project NOAH or Nationwide Operation on assessment of Hazards. According to Casamabre, RFIDs are crucial for NOAH's micro-mapping project that uses Light Detection and Ranging (LiDAR), an equipment designed to survey the entire topography of the Philippines at very high resolution.

According to Aquilan (2004) made a comparable thesis on the automation of time attendance. That records the time in and time out of every employee using barcode system. It tends to eliminate the manual recording system of time and attendance and also include salary computation of each employee based on the time and attendance report.

According to D. Lourdes (2012) from the article entitled “The DOST-ESEP Libraries: The First Library Network in the Philippines” the planning of this system are way back year 1995 but it’s difficult to implement in during those time so it took long time to be able to release it on the public. There some universities that quickly get the partnership since it was started in 2012. It means networking is also a major key on collaborating inside the library. Therefore, this type of system is exactly a big help on the state of the Philippine libraries. It can make our library a world class and can compete on the other country’s libraries.

Ramon Faloran (2005) wrote in the article “The Computer Edge of the New Employment and Opportunities “in the Philippine Daily Inquirer. He stated that computer gives you a different feeling about what is happening in the company. Business will be highly competitive and innovative because the computer provides instant information.

According to Eden (2011), Views that most libraries are forced to work with fewer staff because of the having a more improved and modernized system for their library. According also to the study, the library administration needs to emphasize to their staff that the organization has a vested interest in providing them with the tools and



training they need to assist the organization in the new information marketplace. These may include search engines, circulation systems, and the online public access catalog.

"Biometrics Technology to be Implemented in PNP", 2009 The Philippine National Police (PNP) also made by the international crime detection technologies in the market, the said Philippine agency decided to adopt NEC's (mentioned above) Automated Fingerprint Identification System. Via the technology, PNP staff and experts are able to authenticate unique body demographics, such as fingerprints in the particular case. The technology also allows an almost instantaneous identification. The technology was used however by first building an initial database of 210,000 fingerprints. According to the mentioned above with the use biometrics technology the PNP was able to identify a suspect because of the fingerprints. The city's Mayor, Jun Bernabe, says the new attendance machines will enable it to resolve the issues of long queues as well as employee complaints. A total of 40 Model 628 Double Engine ZK Fingerprint Standalone Time Recorders were purchased. City hall took 15 units while the rest were installed in other local government facilities. The upgrade was jointly run by the city government's Human Resource Management Office and the Electronic Data Processing (EDP) Office. "A side from cutting manpower downtime, it makes timekeeping and payroll preparation simpler, systematic and more accurate. It also provides an additional way to identify, discipline and fire habitual later-comers, absentees and ghost employees," says Mayor Bernabe.



Study by Cantoma (2004) in her thesis entitled “Computer Library System for St. James Academy” stated that, in manual system in retrieving, maintaining security and piling records take place because of the years gone by. Furthermore, these files were only kept in envelopes and folders in wooden racks. There are also instances when the right information is given to a wrong person, which affect quality of service.

Sciences (CHSS) and the College of Science and Mathematics (CSM). The Main Library transferred to the newly renovated building where the main collection was set-up beside the books of the CHSS and the CSM. In July 2002, the library established the Library Educational Media and Information Technology Office (LEMITO). The LEMITO handles the audio-visual services, internet and the connectivity of the library. It currently holds office at the Elias B. Lopez Hall Dormitory. Later, the Main Library collection which grew from 100 titles to 6,191 volumes at the end of May 2004, and was transferred to the Administration Bldg today, the Main Library occupies the former UP in Mindanao Cultural Center (UPMCC) building.

2.2 RELATED STUDIES

2.2.1 FOREIGN STUDIES

ID/Admin number from the SIMS student record. Again, all card readers were connected to a PC. Access to all rooms can be controlled through 'timed' and 'dated' cards. One of the other participating colleges gave complete commitment to a swipe card registration system, only to later discontinue it. In summer 1990 the entire college site



was wired with readers and through 1990-1991 a pilot scheme ran on six courses. This was reasonably effective, mainly because everyone knew it was a pilot and so staff and students were vigilant about working it. Full implementation was introduced in 1991. However, since this was a response to a need for careful monitoring following financial difficulty it was not a strategically planned process. The swipe system was relatively primitive: 'dumb' readers which simply displayed a red light when a swipe was done. Cards were very basic. Lecturers had to swipe two cards to identify themselves and the module or class they were teaching. Students were required to make a single swipe.

RFID tag type of ID card make the process of taking the attendance easier, faster and secure as compared to conventional method. Students or workers only need to place their ID card on the reader and their attendance will be taken immediately. With real time clock capability of the system, attendance taken will be more accurate since the time for the attendance taken will be recorded. The system is connected to the computer through RS232 and store the attendance taken inside database. The proposed system utilized GSM short message service to perform remote data monitoring. The communication software written in VB language achieved efficient control of serial interface port and real-time synchronization of remote data into system database. The computer and GSM (Global System for Mobile communication) Module i.e. SIM 300 are connected by RS232.

Columbia State Community College (2006) uses computer identification account (Log in Name and Password) for access to the institutions computer usage logs occurs electrically through the individual users log in/out process. Users are also informed that they are responsible for any computer file's, database and/or internet sites



that are accessed through their computer identification account, notwithstanding their failure to adhere to the log in/out process, on their voluntary publication of their account information to others.

My Time Force (2006), Time Force is the key to proficient and successful time tracking and management in your workplace. It is a software-based time and attendance system that allows the user to collect and organize employee time data simply and accurately. You can use Time Force to: (a) Easily and efficiently track your employee time; (b) Manage your time & attendance data and employee profiles; (c) Eliminate buddy punching; (d) Make employee scheduling a breeze; (e) Reduce the headaches and time associated with payroll. Best of all, Time Force can quickly pay for itself in time savings, reduction of time theft and elimination of payroll errors. Time Force is the manifestation of 10years of experience, customer feedback, and research and development in the time and attendance market. Over 35,000 businesses use Time Force.

According to Nasir Uddin (2011), for the developing economies like Bangladesh capital market is like an engine for growth. It acts as an intermediary between surplus units and deficits units. Whereas, transparent and investors friendly business atmosphere is essential in capital market for building confidence of general investors. There are a wide range of investors from well-known informed to rumourmonger, the gambler to complete risk avoider. So, a common strategy from the regulator should be taken to balance the whole market. The monitoring investigation, grievance and legal department of DSE plays an important role to build up investor's confidence by providing service to its broker house as well as the investors. The bourse

has a strong monitoring cell, which prime objective is to provide a perfect working environment by providing service to the member houses and investors. Since 2003 the monitoring cell is doing its operation by a fearless team. Over the years, the department has earned its trust and confidence and support through its enthusiasm. This paper is an outcome of study conducted in DSE; the study has identified the reasons for some irregularities in the stock market and tried to find out what are the loopholes to the securities laws and regulations. The findings of the study revealed that the present market is not limited with the high-income people of the society; the small income investors are also spontaneously investing their savings.

According to Selamat and Majlis (2006) in their study entitled 'Challenges in Implementing RFID Tag in a Conventional Library' opined that while implementing RFID technology in IMEN University of Malaysia, main challenges during the implementation related to system integration, parallel operation with existing system, and procedure changes. They highlighted that one of the main challenges was to integrate RFID tagging system with the Library Management System. They opined that self-check counter makes borrowing and returning book process more automated with less involvement of librarians; therefore, librarians can focus on providing more effective work to better serve the library. They suggested that it is necessary to have both systems, i.e., barcode and RFID running in parallel so that library could operate normally while conversion process is in progress.

Bansode and Desale, (2009) discussed that according to their case study entitled 'Implementation of RFID technology in University of Pune Library' highlighted various problems faced by library staff with regard to the lack of availability of technology experts, resistance of the users towards change, lack of standards, etc. They reported major benefits of implementing RFID in stock verification, tracking of shelving of the books and missing books, quick circulation of books, Inventory control, etc. They concluded that RFID technology is a boon for a library, as it requires less manpower to service more books and users.

According to Singh & Midha (2008) in their article entitled 'RFID: A new technology in library management systems' discussed the importance of using RFID in libraries and highlighted its various advantages like speedy circulation operations, self-charging/discharging, highly reliable, high-speed inventory control, automated handling of material, long life of RFID tags, etc. They also discussed the disadvantages of RFID systems like high cost, accessibility to compromise and removal of exposed tags. They pointed out some of the characteristics to be considered while evaluating RFID vendors which include security feature, reader collision, tag memory capacity, anti-collision, standards, etc. They concluded that RFID is increasing in popularity among libraries, as the early adopters of this technology have shown that it makes good economic sense for all types of libraries.

Vasishta, (2009) discussed that according to her paper entitled 'Roadmap for RFID Implementation in Central library, PEC University of Technology' described the basic and optional components required for smooth working of the RFID. She outlined various issues and possible solutions involved in the process of implementing RFID

applications. She concluded that although the RFID technology is quite expensive, yet it has yielded excellent results for many libraries throughout the world. It has the capability of making the management process in a library more convenient and the synergy between the latest technology like RFID and libraries can create wonders resulting in empowerment of both users as well as the librarian.

Kumar et al, (2007). in their paper entitled 'Development of RFID Interface for LMS: Solutions and Techniques' examined various issues related to integration of Library Management Software and RFID. They observed that the SIP2 interface developed for integrating RFID with SOUL is a general solution, which can be adopted for any LMS software with slight modifications. They opined that introduction of such a solution in the academic libraries in the country will boost the modernization activities to turn the library to a fully user-centered automated library.

2.2.2 LOCAL STUDIES

There's still some library system in the Philippines such as Integrated library system (iLib) which is similar to Department of Science and Technology - Engineering and Science Education Project (DOST-ESEP) it is Online Public Access Catalog (OPAC) based system that is created by University of the Philippines-Diliman (UPD). There are few libraries in the Philippines that uses RFID base system. A private school based in Manila has the distinction of being the first in the Philippines to have a high-tech self-service library checkout station based on radio frequency ID (RFID).

The RFID system involves an RF tag, an RF reader and a database. A reader scans the tag with the use computer program and transmits the information to the



computer. The captured information will be processed for intended use. Wireless transmission of identification data rather than by manual transcription increases the quality and speed of data transmission and reception. Presently the library of Lyceum of the Philippines-Laguna is using a traditional system wherein students and employees will have to write their names in the log sheet for their attendance in the library. With the current system, the librarian has a hard time in determining the statistics of the library users. Moreover, it is hassle for the users to log in the logbook and most of the time they forgot to log in. This continuously observed situation led the proponent to investigate and find ways on how to address the need of technology. The application of radio frequency identification would be a good solution.

In the 2005 annual report of the Forest Products Research and Development Institute under the Department of Science and Technology, it is said that, “The computerized database and attendance on Forest Products Research and Development Institute was designed to facilitate the Institute's data entry, processing, computation and maintenance of employee time and attendance transactions. The system allows employees to log in/out as the swipe their IDs through a barcode reader attached to a computer terminal. The processing forms include those for legal holidays, travel order, tardiness, under time, vacation and sick leave. Pertinent Report forms have been designed for easier access to all these data.”

According to Lou (1986), a survey on the computer application in 223 libraries in Metro Manila. The study revealed that they prioritize the three library functions to be computerized as cataloging, indexing, and circulation. However, the plans for automation for the libraries cannot be implemented due to financial incapability of the libraries, and

the lack of funds from the government. Despite of this drawback, the libraries are not turning their backs unto the automation of their libraries.

Study by Cantoma (2004) in her thesis entitled “Computer Library System for St. James Academy” stated that, in manual system in retrieving, maintaining security and piling records take place because of the years gone by. Furthermore, these files were only kept in envelopes and folders in wooden racks. There are also instances when the right information is given to a wrong person, which affect quality of service.

In the advent of technology, communication becomes faster and easier in a tap of a fingertip. In an academic institution per se, communication between parents and school is very challenging and important one. It needs full effort to understand one another. One of the worries of the parents is that if their children are really attending or not in the school during their class period. This is one of the challenges that Lipa City Colleges is confronting, since at present, the school has no way of providing the immediate information to the parents if their children are in or out of the school premises in a particular time. Hence, the researcher conducted this system development study which is considered a new technology which helped the school to solve the problem in monitoring the students’ whereabouts. Specifically, it aimed to 1.) Design a system which will maximize the use of SMS module and RFID technology 2.) Develop a system which will guarantee the privacy and security of information saved in a database. 3.) Determine the benefits of using the SMS Module and RFID technology as a method of communication between the parents and the school 4.) Test and evaluate a system in terms of its performance, functionality, usability, reliability and extensibility.

According to (Q.J. Pastrana, 2013) from the study entitled “Library System with web android based OPAC” technology is a big factor to keep the students keep going to the library. So, the author thinks that android devices can use to attract the students on going to the libraries. In terms of their device they can easily search or monitor the books inside the library. Whether the books that they want to borrow are on the borrowed section of it is still available. So, on this situation the students will have more fun of this new library system. Therefore, the author uses what is the trending device to make students come to the libraries because their task on searching books inside the library become easier and faster.

According to Rhowel M. Dellosa, Elinor Hemedes (2014) The study aimed to develop a computer program that monitored the library users by means of capturing their attendance through radio frequency identification (RFID). A Radio Frequency (RF) reader, RF tag, web camera and computer were utilized to develop and perform the system’s operation. The researcher used developmental method to fulfill its objectives. The system was able to capture the attendance of the library users with the aid of their identification card. The system is acceptable based on the test and evaluation made by the students during system implementation in the library. Several recommendations in connection with the tapping of identification card, the full utilization of capturing device, database update, client-server application, interface design and display of unreturned books.



According to Angela M. Verzosa (1994) In the Philippines, only a few academic and or research libraries have taken the lead in spearheading library automation. As one of the three leading educational institutions, De La Salle University took on the challenge of charting a strategic course towards a totally automated library environment. This case study attempts to make an assessment of the University Library's computerization planning process, through an analysis of the issues and concerns it had to face in developing an automation program and concludes with a strategic plan for its implementation and future directions.

Book monitoring in a library set-up is essential not only for library administrators but for students and faculty as well. Library administrators need to close certain sections of their libraries in order to perform inventory. For students and faculty, the task of locating books in shelves is often disregarded. Searching books in the library can be hard when books that are searched are misplaced or missing from their specified shelves. Borrowing and returning functions of students can also be done without the need for library supervision. Radio Frequency Identification (RFID) Technology used in a library set-up provides solutions to these problems. By applying a program that incorporates the RFID reader with high-frequency tags attached in books, the tasks of library personnel, students and faculty in searching, monitoring and supervising books are supported and made easier.

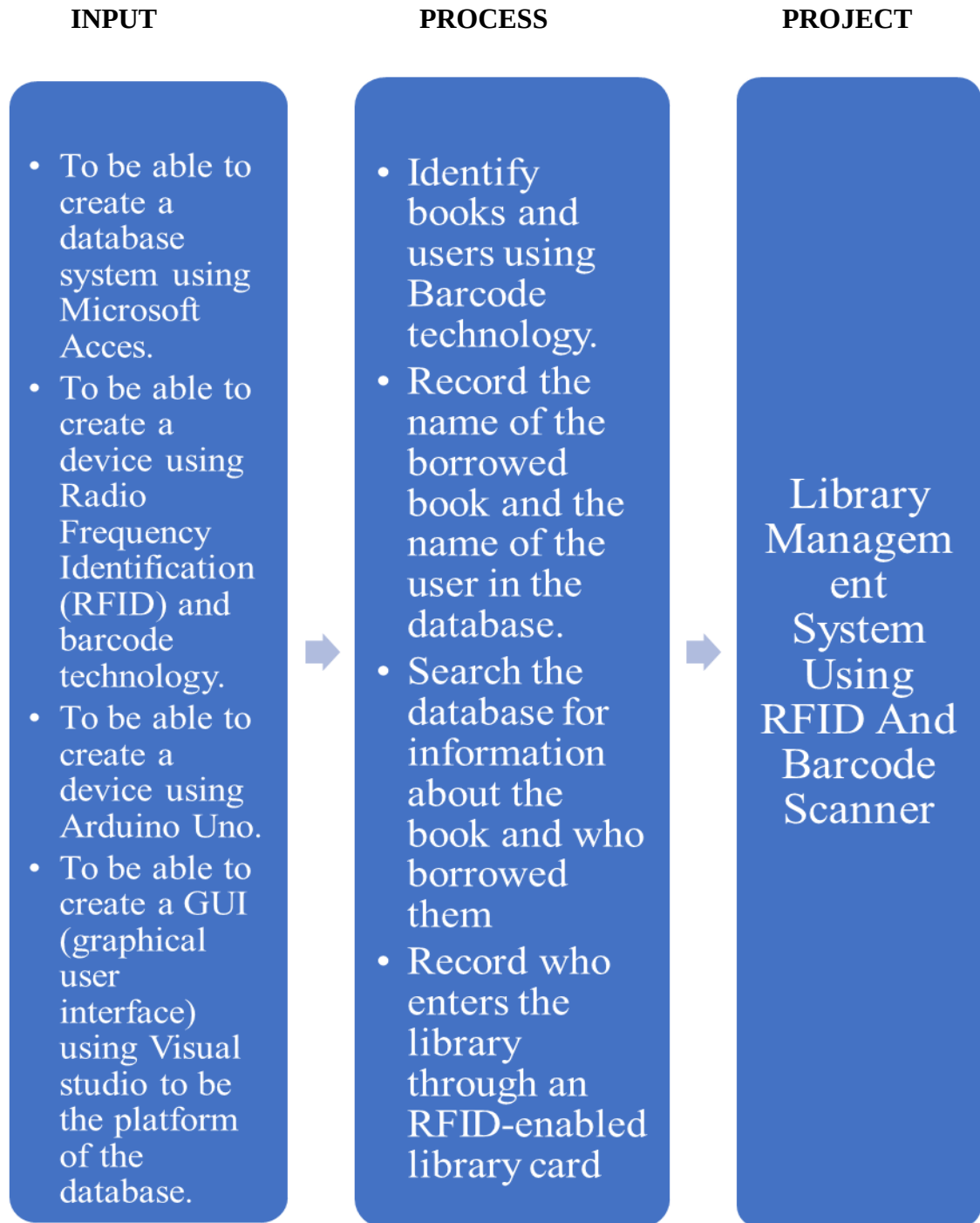
2.3 CONCEPTUAL FRAMEWORK**Fig. 2.1 CONCEPTUAL FRAMEWORK**

Figure 2.1 depicts the objectives of the researchers in the input section. The researchers will use Barcodes and RFID to identify the books and the user respectively. The Barcodes are located inside the books in the library and another RFID tag is inside the user's library card. The barcode will be scanned by a Barcode scanner then sent through the local area network by a computer to the main server. The RFID tag will be read by the RFID reader that is connected to an Arduino UNO. The Arduino will be the one who decodes the RFID information. There will be two RFID terminals, one will scan the user's library card as the user enters the library, and the second one is where the user can scan to borrow or return a book. This data will be sent through a Local Area Network to the main server computer. The computer contains a searchable database where it registers the book and its user. The user can search this database to find more information about the book and if the book is available to be borrowed.

CHAPTER III

METHODOLOGY

This chapter discusses the development of the project to answering the questions in the system. This also provides the outline of the project's methodology, the research and development method, project design, operation flow, descriptions of materials used, project development, testing methods, evaluation process and criteria, instrument and technique used, and the analysis of data gathering methodology.

3.1 PROJECT DEVELOPMENT MODEL

RESEARCH AND DEVELOPMENT METHODS OF RESEARCH

The researchers used Research and Development procedure. This process is a systematic activity which aims to discover solution for existing problems or to create a new goods and knowledge with the use of both basic and applied research. Development is the systematic utilization of the knowledge or understanding gained from research toward the production of useful materials, devices, systems, or methods, including design and development of prototypes and processes. The researchers will be discussing the Design, Fabrication and Evaluation phases in this area.

3.1.1 Design phase

In design phase, the researchers gather all the data and materials in order to provide the objective of the proposed system. The proposed system titled as “BARCODE AND RFID BASED LIBRARY MANAGEMENT SYSTEM” which is a system that uses RFID tags and barcode to record the transactions happens in the library through database. The researchers also make use of Arduino IDE as the microprocessor, Microsoft Access for database and Visual Basic for the GUI (Graphical User Interface) of the system. The hardware of the proposed system composed of Arduino, RFID Tags, Barcode Sticker, Buzzer Server (Computer), Arduino’s RFID reader, Barcode scanner.

3.1.2 FABRICATIONAL PHASE

The researchers make use of barcodes on the books and RFID Tags on the library card of the users. This RFID tag would be a passive RFID tag requiring no power source for it to function.

The researchers also make use of RFID reader. The reader would be connected to an Arduino Uno microcontroller to act as the terminals that read the RFID tags. The Arduino Uno sends the data to a computer, then this computer sends the received from the Arduino through a Local Area Network.

Located in the LAN are two computers. The first computer is running Microsoft Access servers which contain the database for both the books and the users. The second computer is the user-facing computer. This computer terminal is where the users can search for books in the library and to log.

3.1.3 EVALUATION PHASE

To ensure the functionality and quality of the project, the researchers have conducted a self-testing method. This method helps the researchers to know the problems of the project before the evaluation. To be able to test it, the researchers use the project inside the library. They connect their device into the librarian's computer. After the self-testing method, the researchers consulted the students, faculty and expert to check the functionality, reliability, maintainability, extensibility, and economic feasibility of the project, and ask some advices or opinion on how they will enhance and improve their project. Lastly, the researchers conducted an evaluation of the project to some respondents to determine the weaknesses of the project that helps them to improve it.

3.2 PROJECT DESIGN

The project aims to make the use of the library more convenient for both the user and the library staff. The project aims to smooth out the process of borrowing a book from the library.

The project uses RFID tags and Barcodes to keep track of information of the user and the books in the library respectively. Each book in the library would contain a Barcode with a unique identifier for different books. For a user to borrow a book, the user must scan the barcode of the book on a barcode scanner.

The RFID reader is connected to an Arduino Uno Microcontroller that decodes the RFID tag. The Arduino and the barcode scanner in turn sends the data to a computer



then this computer sends the data through a Local Area Network to a server to process the data. The server is running a Microsoft Access database that contains all of the data regarding the books in the library. This data contains the title of the book, author of the book, the subject of the book, and the amount of the available book/s. The server also contains another database which has the data about the users. This data only contains the details of the users who are registered to the library. This means that for each user who wants to use the library, they must avail a library card. The library card is made of RFID tag which is unique to the user. This means the system can provide a way to track who goes in the library by having the user check-in when entering the library. To acquire all the data needed for the system to work, the researchers sets up a terminal where new books and users can register into the system.

The process of borrowing a book requires the user to search for the book that they want to borrow. The project is equipped of another computer terminal connected to the server. This terminal contains a search function and the logging function for students to search for a certain book and to log in and log out to the library.

3.3 OPERATION FLOW

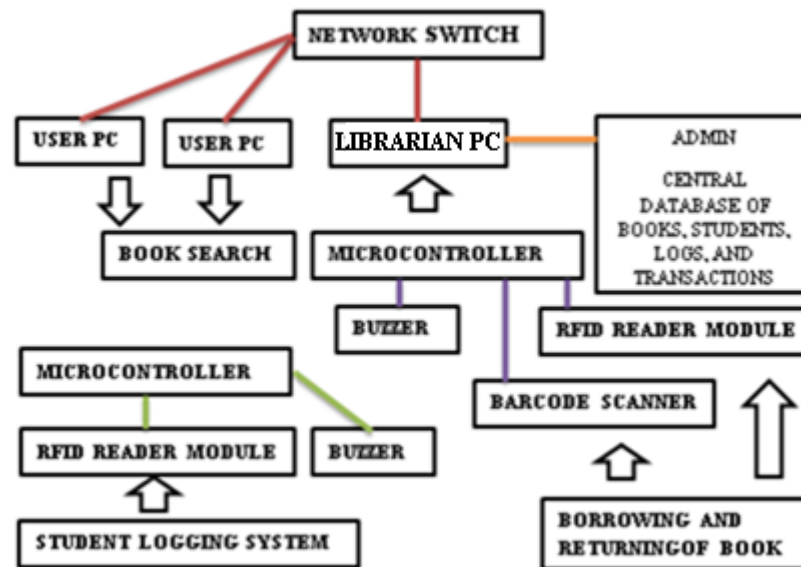


FIGURE 3.1 BLOCK DIAGRAM OF THE SYSTEM

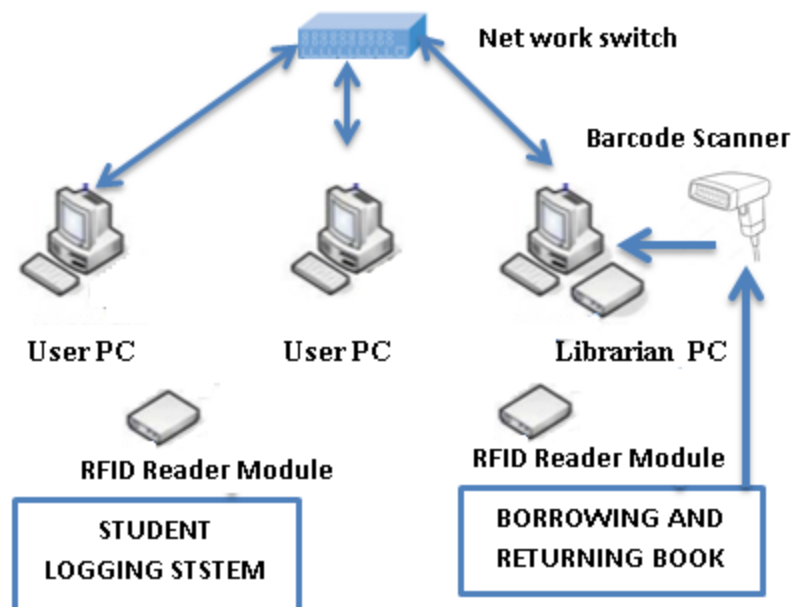


FIGURE 3.2 SYSTEM ARCHITECTURE



Barcode System are used to automate data collection where hand recording is neither timely or cost effective. RFID system uses tags, or labels attached to the objects to be identified. Two-way radio transmitter-receivers called interrogators or readers send a signal to the tag and read its response. RFID Tags are attached in the books and on library cards of the user that sends radio frequency in the RFID Reader through tapping. RFID reader is placed on the booth of the library and will send signal to the Arduino to send data to process the server to a Local Area Network.

3.4 DESCRIPTION AND SPECIFICATIONS OF MATERIALS



FIGURE 3.3 ARDUINO UNO

ARDUINO UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM dountputs), 6 analog iputs, a 16MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller, simply connect it to a computer with a USB cable or power it with AC-to-DC adapter or battery to get started.



FIGURE 3.4 Radio Frequency Identification card

Radio Frequency Identification card uses electromagnetic fields to automatically identify and track tags attached to objects. The tags contain electronically-stored information. Passive tags collect energy from a nearby RFID reader's interrogating radio



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waves. Active tags have a local power source (such as a battery) and may operate hundreds of meters from the RFID reader.



FIGURE 3.5 Radio Frequency Identification Readers

Radio Frequency Identification Reader is a device used to gather information from an RFID tag, which is used to track individual objects. Radio waves are used to transfer data from the tag to a reader.



FIGURE 3.6 BARCODE SCANNER

BARCODE SCANNER is an electronic device that can read and output printed barcodes to a computer. Like a flatbed scanner, it consists of a light source, a lens and a light sensor translating optical impulses into electrical ones. Additionally, nearly all barcode readers contain decoder circuitry analysing the barcode's image data provided by the sensor and sending the barcode's content to the scanner's output port.



FIGURE 3.7 CUSTOMIZE BARCODE STICKER

CUSTOMIZE BARCODE STICKER is ready to meet your barcode labeling needs with pre-printed or custom barcode labels.



FIGURE 3.8 PIEZO ELECTRONIC BUZZER ALARM

PIEZO ELECTRONIC BUZZER ALARM is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers, and confirmation of user input such as a mouse click or keystroke.



FIGURE 3.9 NETWORK SWITCH

NETWORK SWITCH is a computer networking device that connects devices together on a computer network by using packet switching to receive, process, and forward data to destination device. A network switch is a multiport network bridge that uses hardware addresses to process and forward data at the data link layer (layer 2) of the OSI model.

**AMA****COMPUTER COLLEGE OF MALOLOS
COLLEGE OF ENGINEERING****FIGURE 3.10 PERSONAL COMPUTER****Table 3.1 Specification for Personal Computer**

CPU	Intel BayTrail-T CR (Z3735F) Quad Core /1.8GHz
OS	Windows 10
GPU	GEN7 GPU accelerated processor
RAM	DDR3, 1GB
Display	10.1 Inch IPS,1280x800 (WXGA -screen)
Memory Capacity	32GB on board Storage Memory
Support	Micro SD Expansion up to 64GB

PERSONAL COMPUTER is a multi-purpose computer whose size, capabilities, and price make it feasible for individual use. PCs are intended to be operated directly by an end user, rather than by a computer expert or technician. Computer time-sharing models that were typically used with larger, more expensive minicomputer and

mainframe systems, to enable them to be used by many people at the same time, are not used with PCs.

BILL OF MATERIALS

The Table 2 presents the overall material cost of the proposed system spent for its development.

TABLE 3.2 BILL OF MATERIALS

Product Name	Unit Price	Quantity	Total Price
Arduino Uno	Php 240	2	Php 480
RFID Tag	Php 20	4	Php 80
RFID Reader module	Php 250	2	Php 500
BARCODE SCANNER	Php 1,500	1	Php 1,500
PRINTED BARCODE	Php 1	5	Php 5
BUZZER	Php 30	1	Php 30
Total Cost	-	-	Php 2595

3.5 PROJECT DEVELOPMENT

TABLE 3.3 The Researchers' Calendar of Activities Using Gantt Chart

Activities	June				July					August				September				
	Wk 1	Wk 2	Wk 3	Wk 4	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 1	Wk 2	Wk 3	Wk 4	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5
Conducting of Research																		
Title Presentation																		
Title Approval																		
Preparation of Chapter 1																		
Submission and Checking of Chapter 1																		
Preparation of Chapter 2																		
Submission and Checking of Chapter 2																		
Preparation of Chapter 3																		
Checking of Chapter 3																		
Title Defense																		

The researchers conducted their study on the beginning of the First Trimester, S.Y. 2018-2019, starting from the third week of June. After a week, the researchers presented ten titles for approval. The approved title was given the following week. In the second week of July, the preparation for the first chapter was underway. This includes research, and discussions for Chapter 1. The manuscript of Chapter 1 was submitted on the third week of July and was resubmitted the following week after some revisions. That same week the preparation of Chapter 2 was started. Chapter 2 was submitted on the last week of July and then resubmitted after revisions to Chapter 2 the following week. Chapter 3's preparation was started on the first week of August and was submitted the following week. The researchers submitted and defended the project on the first week of September.

3.6 TESTING METHODS

To ensure that the proposed Smart Library System will work properly, the researchers conducted different testing, trials, method and troubleshooting after the development of the system.

1. The researchers inspect the cable connections for each device and assure that the system is working properly.
2. The researchers checks that the components are all in working order.
3. The researchers inspected the synchronization of the components to be sure that the system is operating effectively and efficiently.
4. The researcher checks the database program does not have any bugs.
5. The researchers also asks ten faculty, thirty-five students and five experts to evaluate the project if it is functional and ensure its quality.

3.7 EVALUATION PROCESS

To be able to check the quality of the system, the researcher begun an evaluation process. The respondents are issued with an evaluation sheet containing the criteria during the evaluation. The researchers begin with presenting their project before the

evaluation. The following are the 50 individuals that the researchers ask to evaluate for the quality test of this project.

A. Students (35)

B. Faculty (10)

C. Experts (5)

3.8 EVALUATION CRITERIA

The Table 4 presents the criteria that the respondents observed and considered during the evaluation of the system.

TABLE 3.4 CRITERIA FOR SYSTEM EVALUATION

Criteria	Indicator
Functionality	• The proposed system works properly and all the expected outcomes are delivered. • The system is user-friendly and has met the work requirements of the users.
Reliability	• The proposed system can be used in any time and place with minimal or without consequences in its functionality. • The system performs with minimal or without errors during its operation.
Maintainability	• The proposed system requires only minimal maintenance after its operation.

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Extensibility	• The system supports adaptability, coping with the needs of the end-users. • The system can still be integrated with other electronic devices in the future.
Economic feasibility	• The proposed system offers long-term usage and can be considered as a good investment. • The total cost of the system is practically worth of its performance.

3.9 INSTRUMENT AND TECHNIQUE USED

The researchers used an evaluation sheet contained all the necessary criteria for the evaluation of the system.

TABLE 3.5 EVALUATION SHEET FOR THE PROPOSED SYTEM

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City of Malolos, Bulacan

SOLAR POWERED COIN BASED MOBILE CHARGING STATION

EVALUATION SHEET

Name of Respondent: _____

Date: _____

Occupation: _____ Age: ____ Sex: _____

INSTRUCTION: Rate the system based on the given criteria by encircling the corresponding number.

EVALUATION CRITERIA	Excellent	Very Satisfactory	Satisfactory	Fair	Poor
FUNCTIONALITY					
1. The proposed system works properly and all the expected outcomes are delivered.	5	4	3	2	1
2. The system is user-friendly and has met the work requirements of the users.	5	4	3	2	1
RELIABILITY					
1. The proposed system can be used in any time and place with minimal or without consequences in its functionality.	5	4	3	2	1
2. The system performs with minimal or without errors during its operation.	5	4	3	2	1
MAINTAINABILITY					
1. The proposed system requires only minimal maintenance after its operation.	5	4	3	2	1
2. The integrated devices are carefully placed and is rugged in construction.	5	4	3	2	1
EXTENSIBILITY					
1. The system supports adaptability, and can cope with the varying needs of the end-users.	5	4	3	2	1
2. The system can still be integrated with other electronic devices in the future.	5	4	3	2	1
ECONOMIC FEASIBILITY					
1. The proposed system offers long-term usage and can be considered as a good investment.	5	4	3	2	1
2. The total cost of the system is practically worth of its performance.	5	4	3	2	1

COMMENTS AND RECOMMENDATIONS:

EVALUATOR'S SIGNATURE

3.10 ANALYSIS OF DATA GATHERING METHODOLOGY

After the evaluation of the responders in the evaluation process, the researchers tallied the data that they gathered to them using the Likert Scale as seen in the Table 3.6. A Likert item is simply a statement which the respondent is asked to evaluate according to any kind of subjective or objective criteria; generally, the level of agreement or disagreement is measured. This can help the researchers to determine the capability. The data has to be thoroughly checked to fulfill the strict formal axioms of the project.

TABLE 3.6 The Likert Scale

Rating	Scale	Range
5	Outstanding	4.51-5.0
4	Above Satisfactory	3.51-4.5
3	Satisfactory	2.51-3.5
2	Below Satisfactory	1.51-2.5
1	Needs Improvement	1.0-1.50

CHAPTER IV

TESTING, EVALUATION AND RESULTS

This chapter presents the testing and evaluation of the system, including the computation and analysis of gathered data as well as the interpretation of results.

4.1 Initial testing stages of the project

The researchers conducted troubleshooting after the hardware was integrated and an initial testing of the device and its software to eliminate bugs. This was done to check if the system is working properly and efficiently. The researchers have thoroughly checked the wire connection and correct positioning and installation of each component inside the device to ensure all parts are functioning properly.

4.2 Evaluation of the system

The researchers have conducted an evaluation to assess the acceptability and feasibility of the system with the help of various professionals from different fields. The researchers have used the stratified sampling method to obtain respondents. The respondents were composed of 10 professional, 35 students, and 5 experts. The respondents were asked to assess the potential of the development of the study in their area. The researchers provided them with evaluation sheets containing all the necessary criteria for the evaluation of the system.

The respondents have evaluated the system for its functionality, usability, performance and maintenance, which were explained earlier in the Evaluation Criteria section of Chapter 3.

The computation of the weighted means was done in order to interpret the results and presented by tables.

The following tables are the evaluation result of the weighted means.

Table 4.1 Evaluation on Functionality

Professional	4.7	Outstanding
Expert	4.6	Outstanding
Student	4.6	Outstanding
Mean Score	4.63	Outstanding

Table 4.1 presents the evaluation on functionality of the system. Static shows that the functionality of the system is rated “Outstanding” with the computed mean score of 4.62.

This data imply that the respondents are astounded by the functionality of the project. The system works properly, and all the expected outcomes are delivered. The system is user friendly and has met the work requirements of the users.

Table 4.2 Evaluation on Reliability

Professional	4.65	Outstanding
Expert	4.7	Outstanding
Student	4.3	Above Satisfactory
Mean Score	4.55	Outstanding

Table 4.2 presents the evaluation on reliability of the system. Static shows that the reliability of the system is rated “Outstanding” with the computed mean score of 4.55.

This data imply that the respondents are astounded by the reliability of the project.

It can be used in any time and place with minimal or without consequences in its functionality. The system performs with minimal or without errors during its operation.

Table 4.3 Evaluation on Maintainability

Professional	4.4	Above Satisfactory
Expert	4.4	Above Satisfactory
Student	4.4	Above Satisfactory
Mean Score	4.4	Above Satisfactory



Table 4.3 presents the evaluation on maintainability of the system. Static shows that the maintainability of the system is rated “Above Satisfactory” with the computed mean score of 4.4.

This data imply that the respondents are above satisfied in maintainability of the project. The system only requires only minimal maintenance after its operation. The integrated devices are carefully placed inside the case and the system is rugged in construction.

Table 4.4 Evaluation on Extensibility

Professional	4.65	Outstanding
Expert	4.5	Above Satisfactory
Student	4.53	Outstanding
Mean Score	4.56	Outstanding

Table 4.4 presents the evaluation on extensibility of the system. Static shows that the extensibility of the system is rated “Outstanding” with the computed mean score of 4.56.

This data imply that the respondents are astounded by the extensibility of the project. The system supports adaptability, coping with the needs of end-user. The system can still be integrated with other electronics devices in the future.

**Table 4.5 Evaluation on Economic Feasibility**

Professional	4.7	Outstanding
Expert	4.8	Outstanding
Student	4.5	Above Satisfactory
Mean Score	4.67	Outstanding

Table 4.5 presents the evaluation on feasibility of the system. Static shows that the feasibility of the system is rated “Outstanding” with the computed mean score of **4.67**.

This data imply that the respondents are astounded by the feasibility of the project. The project offers long term usage and can be considered as a good investment. The total cost of the system is practically worth of its performance.

4.3 Overall Evaluation

As stated in Chapter 3, the researchers have used the Likert scale to measure and interpret the evaluation results of the respondents. Based on the results, the respondents agree that the system is highly functional, maintainable usable and performs well for the blind person.

**AMA****COMPUTER COLLEGE OF MALOLOS
COLLEGE OF ENGINEERING****Table 4.6 Summary of Overall Evaluation**

Functionality	4.63	Outstanding
Reliability	4.55	Outstanding
Maintainability	4.56	Outstanding
Extensibility	4.4	Above Satisfactory
Economic Feasibility	4.67	Outstanding
Grand Mean	4.562	Outstanding

Table 4.6 presents the Summary of Overall Evaluation of the system. Static shows that the feasibility of the system is rated “Satisfactory” with the computed mean score of 3.76.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 SUMMARY

The researchers underwent various steps in the development of Library Management System. The need to minimize work of librarian and to reduce processing time of borrowing and returning of books also the logging of the students, the researchers develop a RFID and barcode technology-based device that intends to address these concerns.

Initially, the researchers conducted research and collected significant ideas from related literatures and studies regarding the efficiency and operation of RFID and Barcode. Next, the researchers carefully planned the system's design and identified the components that were used in the system's construction.

After the analysis and conceptualization of the system design, the researchers bought all the necessary components and tools then proceeded to the construction of the circuitry and the integration of the system's components.

The researchers did thorough tests after its construction for output analysis comparison and made adjustments to correct and maintain the proper operation of the system. Lastly, the researchers conducted an evaluation from a gathered group of respondents to determine the acceptability and feasibility of the system.

All of the stated processes have eventually led to the output: "Library Management System Using RFID and Barcode Scanner".

5.2 CONCLUSION

After painstaking work and thorough experimentations, the researchers can say that the objectives of this particular study have been successfully achieved. The study Library Management System Using RFID and Barcode has been successfully achieved. The overall functionality, reliability, maintainability, extensibility, economic feasibility of the research based on the respondents is satisfactory. The overall evaluation of the research is satisfactory based on the Likert scale with a grand mean of 3.73. this is a study that intends to reduce the process of borrowing and returning of books and the logging of the students using RFID as the library card of each students and Barcode as the id of each books, as well as to log all the transactions and the status of the transaction whether who borrows a certain book, when the book is borrowed, when the book returned, identifying penalties whether the book returned before the due date and the catalogue of the book containing the definition, status, and the shelf number of the book through a database using Microsoft Access and visual basic as the GUI, the transactions can be seen in the librarians computer as the server and the students can search a book in a client computer connected into LAN. After careful and thorough analysis and evaluation of the results, the researchers firmly believe that the project have met the objectives in producing device and provide significant benefits to the society.

RECOMMENDATION

The project study can be further improved by overcoming its limitations. Several recommended modifications that can improve the performance of the project are the following:

- All scanners should be in the server only
- Make use of raspberry pi instead of arduino to make the device portable
- Make use of the tracking ability of RFID
- Use RFID for admins account



APPENDICES

A. SYSTEM CODE

A.1 AddBooks Class

```
1      Public Class AddBooks
2      Public NameFrm, NameTo As String
3      Private Sub Metrobutton9_Click(ByVal sender As System.Object, ByVal e
      As System.EventArgs) Handles MetroButton9.Click
4
5          LibraryMainForm.Show()
6          Me.Close()
7      End Sub
8
9      Sub Firstload()
10         MetroTextBox1.Focus()
11
12 End Sub
13
14     Private Sub AddBooks_Load(ByVal sender As System.Object, ByVal e
      As System.EventArgs) Handles MyBase.Load
15         'Call generateyear()
16         Call disablethem()
17         Call readData()
18         Call Name_Group()
19         MetroTextBox1.Select()
20         Call fill_list()
21
22
23
24
25 End Sub
26
51 Private Sub MetroComboBox1_LostFocus(ByVal sender As Object,
      ByVal e As System.EventArgs) Handles MetroComboBox1.LostFocus
52     MetroComboBox1.Text = MetroComboBox1.Text.ToUpper()
53 End Sub
54
55
56
```



```
57 Private Sub Metrobutton1_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MetroButton1.Click
58
59     Call enablethem()
60     MetroTextBox1.Select()
61 End Sub
62
63 Private Sub MetroTextBox2_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs) Handles MetroTextBox2.LostFocus
64     NameFrm = MetroTextBox2.Text
65     Call Sentence()
66     MetroTextBox2.Text = NameTo
67 End Sub
68 Sub disablethem()
69
70
71     MetroTextBox2.Enabled = False
72     MetroTextBox3.Enabled = False
73     MetroComboBox1.Enabled = False
74     MetroTextBox4.Enabled = False
75     MetroTextBox5.Enabled = False
76     MetroTextBox6.Enabled = False
77     MetroTextBox8.Enabled = False
78
79
80 End Sub
81 Sub enablethem()
82
83     MetroTextBox2.Enabled = True
84     MetroTextBox3.Enabled = True
85     MetroComboBox1.Enabled = True
86     MetroTextBox4.Enabled = True
87     MetroTextBox5.Enabled = True
88     MetroTextBox6.Enabled = True
89     MetroTextBox8.Enabled = True
90
91     MetroTextBox1.Text = ""
92     MetroTextBox2.Text = ""
93     MetroTextBox3.Text = ""
```



```
94 MetroTextBox4.Text = ""
95 MetroTextBox5.Text = ""
96 MetroTextBox6.Text = ""
97 MetroTextBox8.Text = ""
98 MetroComboBox1.SelectedIndex = -1 'clearing combo box
99 'MetroTextBox8.SelectedIndex = -1 'clearing combo box
100
101
102 End Sub
103
104 Sub Sentence()
105 Dim a, b As Integer
106 a = NameFrm.Length
107 NameTo = ""
108 For b = 0 To a - 1
109 If b = 0 Then
110 If Char.IsLower(NameFrm(0)) Then
111 NameTo = Char.ToUpper(NameFrm(0))
112 Else
113 NameTo = NameFrm(0)
114 End If
115 Else
116 If NameFrm(b - 1) = " " Then
117 NameTo = NameTo + Char.ToUpper(NameFrm(b))
118 Else
119 NameTo = NameTo + NameFrm(b)
120 End If
121 End If
122 Next
123 End Sub
124
125 Private Sub MetroTextBox3_LostFocus(ByVal sender As Object, ByVal e
As System.EventArgs) Handles MetroTextBox3.LostFocus
126 NameFrm = MetroTextBox3.Text
127 Call Sentence()
128 MetroTextBox3.Text = NameTo
129 End Sub
130
```



```
131 Private Sub MetroTextBox3_TextChanged(ByVal sender As System.Object,  
    ByVal e As System.EventArgs) Handles MetroTextBox3.TextChanged  
132  
133 End Sub  
134  
135 Private Sub MetroTextBox4_LostFocus(ByVal sender As Object, ByVal e  
    As System.EventArgs) Handles MetroTextBox4.LostFocus  
136 NameFrm = MetroTextBox4.Text  
137 Call Sentence()  
138 MetroTextBox4.Text = NameTo  
139 End Sub  
140  
141 Private Sub MetroTextBox4_TextChanged(ByVal sender As System.Object,  
    ByVal e As System.EventArgs) Handles MetroTextBox4.TextChanged  
142  
143 End Sub  
144  
145 Private Sub MetroTextBox5_LostFocus(ByVal sender As Object,  
    ByVal e As System.EventArgs) Handles MetroTextBox5.LostFocus  
146 NameFrm = MetroTextBox5.Text  
147 Call Sentence()  
148 MetroTextBox5.Text = NameTo  
149 End Sub  
150  
151  
152  
153  
154 Public Sub adddatatolistview(ByVal lvw As ListView, ByVal BookID As  
    String, ByVal GroupID As String, ByVal BookName As String, ByVal  
    Publisher As String, ByVal Author As String, ByVal PubYear As String,  
    ByVal edi As String, ByVal qty As String) 155 Dim lv As New ListViewItem  
156 lvw.Items.Add(lv)  
157 lv.Text = BookID  
158 lv.SubItems.Add(GroupID)  
159 lv.SubItems.Add(BookName)  
160 lv.SubItems.Add(Publisher)  
161 lv.SubItems.Add(Author)  
162 lv.SubItems.Add(PubYear)  
163 lv.SubItems.Add(edi)
```



```
164         lv.SubItems.Add(qty)
165
166     End Sub
167
168     Private Sub Metrobutton8_Click(ByVal sender As System.Object,
169     ByVal e As System.EventArgs) Handles MetroButton8.Click
170     If MetroTextBox9.Text = "" Then
171     MessageBox.Show("Invalid Book Copy")
172
173     Else
174
175     Try
176
177
178
179     If objcon.State = ConnectionState.Closed Then objcon.Open()
180     If MessageBox.Show("Do you really want to delete?", "ARE YOU
181     SURE", MessageBoxButtons.YesNo) =
182     Windows.Forms.DialogResult.Yes Then
183
184     com = New OleDb.OleDbCommand("INSERT INTO
185     ArchivesBooks VALUES('" & MetroTextBox1.Text &
186     "','" &
187     MetroComboBox1.Text & "','" & MetroTextBox2.Text
188     & "','" &
189     MetroTextBox3.Text & "','" & MetroTextBox4.Text & "','" &
190     MetroTextBox8.Text & "','" & MetroTextBox5.Text & "','"
191     & MetroTextBox6.Text & "','" & MetroTextBox7.Text
192     & "','"
193     & MetroTextBox9.Text & "')" , objcon)
194     com.ExecuteNonQuery()
195     com2 = New OleDb.OleDbCommand("UPDATE Books SET
196     Quantity='" & MetroTextBox6.Text - 1 & " WHERE
197     BookID='"
198     & MetroTextBox1.Text & "'", objcon)
199     com2.ExecuteNonQuery()
```



```
189         If MetroTextBox6.Text = 1 Then
190             com3 = New OleDb.OleDbCommand("DELETE
FROM Books WHERE BookID=" &
MetroTextBox1.Text & "", objcon)
191             com3.ExecuteNonQuery()
192             End If
193
194
195
196             Call readData()
197             Call enablethem()
198             MsgBox("Deleted successfully", 0, "SUCCESS") 199
229         End If
230
231
232
233
234         Catch ex As Exception
235
236         End Try
237
238     End If
239
240     End Sub
241     Sub fill_list()
242         If objcon.State = ConnectionState.Closed Then objcon.Open() 243         com =
New OleDb.OleDbCommand("Select * from Books", objcon)
244         Dim dr As OleDb.OleDbDataReader
245         dr = com.ExecuteReader
246         dr.Read()
247         While (dr.NextResult)
248
249         End While
250     End Sub
251
252     Private Sub MetroTextBox1_KeyPress(sender As Object, e As
KeyPressEventArgs) Handles MetroTextBox1.KeyPress
253         'condition for entering number only
```



```
254      If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
      IsNumeric(e.KeyChar) Then
255      MessageBox.Show("Please enter numbers only")
256      e.Handled = True
257      End If
258      End Sub
259
260      Private Sub MetroTextBox5_KeyPress(sender As Object, e As
      KeyPressEventArgs) Handles MetroTextBox5.KeyPress
261      'condition for entering number only
262      If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
      IsNumeric(e.KeyChar) Then
263      MessageBox.Show("Please enter numbers only")
264      e.Handled = True
265      End If
266      End Sub
267
268      Private Sub MetroTextBox1_TextChanged(ByVal sender
      As System.Object, ByVal e As System.EventArgs)
      Handles MetroTextBox1.TextChanged
269      Dim i As Integer
270      ListView2.SelectedItems.Clear()
271      MetroTextBox1.Focus()
272      Try
273      If Me.MetroTextBox1.Text = "" Then
274      MetroTextBox2.Text = ""
275      Else
276      For i = 0 To ListView2.Items.Count - 1
277      If MetroTextBox1.Text =
      ListView2.Items(i).SubItems
      (0).Text Then
278      MetroComboBox1.Text =
      ListView2.Items(i).SubItems
      (1).Text
279      MetroTextBox2.Text =
      ListView2.Items(i).SubItems (2).Text
280      MetroTextBox3.Text =
      ListView2.Items(i).SubItems (3).Text
```



```
281 MetroTextBox4.Text = ListView2.Items(i).SubItems
    (4).Text
282 MetroTextBox8.Text = ListView2.Items(i).SubItems
    (5).Text
283 MetroTextBox5.Text = ListView2.Items(i).SubItems
    (6).Text
284 MetroTextBox6.Text = ListView2.Items(i).SubItems
    (7).Text
285 ListView2.Items(i).Selected = True
286 Exit For
287 End If
288 Next
289 End If
290 Catch
291
292 End Try
293 End Sub
294
295 Private Sub Button6_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs)
296 Try
297 Dim i As Integer
298 For i = 0 To ListView2.Items.Count - 1
299 If ListView2.Items(i).Selected = True Then
300 MetroTextBox1.Text = ListView2.Items(i + 1).SubItems
    (0).Text
301 Exit For
302 End If
303 Next
304 ListView2.Focus()
305 ListView2.FullRowSelect = True
306 Catch ex As Exception
307
308 End Try
309 End Sub
310
311
312
313 Sub Name_Group()
```



```
314 Try
315 If objcon.State = ConnectionState.Closed Then objcon.Open() 316
    com = New OleDb.OleDbCommand("Select GroupName from
    GroupName", objcon)
317     dr = com.ExecuteReader
318     While dr.Read
319         MetroComboBox1.Items.Add(dr.Item(0))
320     End While
321     dr.Close()
322     objcon.Close()
323     Catch ex As Exception
324
325 End Try
326 End Sub
327
358
359 Private Sub Button5_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs)
360 Try
361     Dim i As Integer
362     For i = 0 To ListView2.Items.Count - 1
363         If ListView2.Items(i).Selected = True Then
364             MetroTextBox1.Text = ListView2.Items(i - 1).SubItems
                (0).Text
365             Exit For
366         End If
367     Next
368     ListView2.Focus()
369     ListView2.FullRowSelect = True
370     Catch ex As Exception
371
372 End Try
373 End Sub
374
375
376
377
378 Private Sub ListView2_SelectedIndexChanged_1(sender As Object, e As
    EventArgs) Handles ListView2.SelectedIndexChanged
```



```
379 '////////////////////////////////////
380 If objcon.State = ConnectionState.Closed Then objcon.Open() 381
382 com = New OleDb.OleDbCommand("SELECT * FROM ArchivesBooks",
    objcon)
383 com.Connection = objcon
384 Dim maxid As Object
385 Dim Strid As String
386 Dim intID As Integer
387
388 com.CommandText = "Select Max(Copy) as maxid from ArchivesBooks
Where BookID='" & MetroTextBox1.Text & "'" 389 maxid = com.ExecuteScalar
390
391 If maxid Is DBNull.Value Then
392     intID = 1
393 Else
394     Strid = CType(maxid, String)
395     intID = CType(Strid, String)
396     intID = intID + 1
397
398
399 End If
400
401 MetroTextBox9.Text = intID
402 objcon.Close()
403
404
405 '////////////////////////////////////
406
407
408 Dim i As Integer
409 For i = 0 To ListView2.Items.Count - 1
410     If ListView2.Items(i).Selected = True Then
411         Call disablethem()
412         MetroTextBox1.Text =
            ListView2.Items(i).SubItems(0).Text413
414     Exit For
415     End If
416 Next
417 ListView2.Focus()
```



```
418         ListView2.FullRowSelect = True
419     End Sub
420
421
422     Private Sub MetroButton3_Click(sender As Object, e As EventArgs)
423         Handles MetroButton3.Click
424         If MetroTextBox1.Text = "" Then
425             MsgBox("Please enter the Book ID!", 0, "")
426         ElseIf MetroComboBox1.SelectedIndex = -1 Then
427             MsgBox("Please select group ID", 0, "") ' Validation (Robert)
428         ElseIf MetroTextBox2.Text = "" Then
429             MsgBox("Please select Book Name", 0, "") ' Validation (Robert)
430         ElseIf MetroTextBox3.Text = "" Then
431             MsgBox("Please select Publisher", 0, "") ' Validation (Robert)
432         ElseIf MetroTextBox4.Text = "" Then
433             MsgBox("Please select Author", 0, "") ' Validation (Robert)
434         ElseIf MetroTextBox8.Text > 2018 Or MetroTextBox8.Text < 1950
435             Then
436                 MsgBox("Invalid Date, Must be less than 2018 and greater than
437                     1950", 0, "") ' Validation (Robert)
438
439         ElseIf MetroTextBox5.Text = "" Then
440             MsgBox("Please select Edition", 0, "") ' Validation (Robert)
441         ElseIf MetroTextBox6.Text = "" Then
442             MsgBox("Please select Edition", 0, "") ' Validation (Robert)
443         ElseIf MetroTextBox6.Text > 99 Then
444             MsgBox("Maximum of 99 quantity", 0, "") ' Validation (Robert)
445         Else
446             Try
447                 If objcon.State = ConnectionState.Closed Then objcon.Open()
448                 com = New OleDb.OleDbCommand("INSERT INTO Books VALUES(""
```



```
449      Call readData()
450      Call enablethem()
451      MsgBox("Saved successfully", 0, "SUCCESS")
452      objcon.Close()
453      Catch ex As Exception
454      MsgBox(ex.Message, 0, "")
455      End Try
456      End If
457      End Sub
458      Sub readData()
459      ListView2.Clear()
460      ListView2.Columns.Add("Book Barcode", 160,
461      HorizontalAlignment.Left)
462      ListView2.Columns.Add("Group ID", 80,
463      HorizontalAlignment.Center)
464      ListView2.Columns.Add("Book Name", 290,
465      HorizontalAlignment.Center)
466      ListView2.Columns.Add("Publisher", 90,
467      HorizontalAlignment.Center)
468      ListView2.Columns.Add("Author", 90,
469      HorizontalAlignment.Center)
470      ListView2.Columns.Add("Publishing Year", 190,
471      HorizontalAlignment.Center)
472      ListView2.Columns.Add("Edition", 90,
473      HorizontalAlignment.Center)
474      ListView2.Columns.Add("Quantity", 100,
475      HorizontalAlignment.Center)
476      ListView2.View = View.Details
477      Try
478
479      If (objcon.State = ConnectionState.Closed) Then objcon.Open()
480      com = New OleDb.OleDbCommand("SELECT * FROM Books",
481      objcon)
482      dr = com.ExecuteReader
483      While dr.Read()
484      Call adddatatolistview(ListView2, dr(0), dr(1), dr(2), dr(3), dr(4),
485      dr(5), dr(6), dr(7))
486      End While
487      dr.Close()
```



```
478         objcon.Close()
479         Catch
480         'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
481         End Try
482         End Sub
483
484
485
486
487
488
489
490     Private Sub MetroTextBox6_KeyPress(sender As Object, e As
KeyPressEventArgs) Handles MetroTextBox6.KeyPress 491
'condition for entering number only
492         If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
IsNumeric(e.KeyChar) Then
493             MessageBox.Show("Please enter numbers only")
494             e.Handled = True
495             End If
496             End Sub
497
500         Private Sub MetroTextBox8_KeyPress(sender As Object, e As
KeyPressEventArgs) Handles MetroTextBox8.KeyPress
501             If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
IsNumeric(e.KeyChar) Then
502                 MessageBox.Show("Please enter numbers only")
503                 e.Handled = True
504                 End If
505                 End Sub
506
507     Private Sub MetroTabPage1_Click(sender As Object, e As EventArgs) Handles
MetroTabPage1.Click
508
509     End Sub
510
511     Private Sub MetroTextBox1_Click(sender As Object, e As EventArgs) Handles
MetroTextBox1.Click
512
```



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513 End Sub
514 End Class



A.2 AddBorrower Class

```
3 Imports System
4 Imports System.IO.Ports
5
6 Public Class AddBorrower
7     Dim receivedData As String = ""
8
9     Function ReceiveSerialData() As String
10         receivedData = ""
11         Dim Reader As String
12         Try
13
14             Reader = SerialPort2.ReadExisting()
15             If Reader Is Nothing Then
16                 Return "Nothing" & vbCrLf
17             Else
18                 Return Reader
19             End If
20             Catch ex As TimeoutException
21                 Return "Error"
22             End Try
23
24
25 End Function
26
27 Function ArduinoConnect() As String
28     Timer2.Enabled = False
29
30     For Each sp As String In My.Computer.Ports.SerialPortNames
31         SerialPort2.Close()
32         SerialPort2.PortName = sp
33         SerialPort2.BaudRate = 9600
34         SerialPort2.DataBits = 8
35         SerialPort2.Parity = Parity.None
36         SerialPort2.StopBits = StopBits.One
37         SerialPort2.Handshake = Handshake.None
38         SerialPort2.Encoding = System.Text.Encoding.Default 'very
39         SerialPort2.ReadTimeout = 10000
```



```
40         SerialPort2.Open()
41         Timer2.Enabled = True
42         Next
43         End Function
44
45
46     Private Sub AddCustomer_Load(ByVal sender As System.Object, ByVal e
47     As System.EventArgs) Handles MyBase.Load
48         MetroDateTime1.MinDate = DateTime.Today
49         Call DisableThem()
50         Call readData()
51         MetroTextBox1.Focus()
52         ArduinoConnect()
53         End Sub
54
55
56     Private Sub Timer2_Tick(sender As Object, e As EventArgs) Handles
57     Timer2.Tick
58         receivedData = ReceiveSerialData()
59         MetroTextBox1.Text &= receivedData
60         End Sub
61     Public NameFrm, NameTo As String
62
63     Private Sub MetroTextBox4_KeyDown(ByVal sender As Object, ByVal e
64     As System.Windows.Forms.KeyEventArgs) Handles MetroTextBox4.KeyDown
65         End Sub
66     Private Sub MetroTextBox4_LostFocus(ByVal sender As Object, ByVal e
67     As System.EventArgs) Handles MetroTextBox4.LostFocus
68         MetroTextBox4.Text = MetroTextBox4.Text.Trim
69         End Sub
70
71     Private Sub MetroButton9_Click(ByVal sender As System.Object, ByVal e
72     As System.EventArgs) Handles MetroButton9.Click
73         SerialPort2.Close()
74         LibraryMainForm.Show()
```



```
74      Me.Close()
75      End Sub
76
77      Private Sub MetroTextBox2_LostFocus(ByVal sender As Object, ByVal e
78      As System.EventArgs) Handles MetroTextBox2.LostFocus
79      NameFrm = MetroTextBox2.Text
80      Call Sentence()
81      MetroTextBox2.Text = NameTo
82      End Sub
83  Private Sub MetroTextBox2_TextChanged(ByVal sender As System.Object,
84  ByVal e As System.EventArgs) Handles MetroTextBox2.TextChanged
85
86      End Sub
87      Sub Sentence()
88      Dim a, b As Integer
89      a = NameFrm.Length      NameTo = ""
90      For b = 0 To a - 1
91      If b = 0 Then
92      If Char.IsLower(NameFrm(0)) Then
93      NameTo = Char.ToUpper(NameFrm(0))
94      Else
95      NameTo = NameFrm(0)
96      End If
97      Else
98      If NameFrm(b - 1) = " " Then
99      NameTo = NameTo + Char.ToUpper(NameFrm(b))
100     Else
101     NameTo = NameTo + NameFrm(b)
102     End If
103     End If
104     Next
105     End Sub
106
107     Private Sub MetroTextBox3_LostFocus(ByVal sender As Object, ByVal e
108     As System.EventArgs) Handles MetroTextBox3.LostFocus
109     NameFrm = MetroTextBox3.Text
110     Call Sentence()
111     MetroTextBox3.Text = NameTo
```



```
111         End Sub
112
113
114
115         Private Sub MetroButton2_Click(ByVal sender As System.Object,
116         ByVal e As System.EventArgs) Handles MetroButton2.Click
117         If MetroTextBox1.Text = "" Then
118             MsgBox("Please enter Borrower's RFID", 0, "")
119         ElseIf MetroComboBox1.SelectedIndex = -1 Then
120             MsgBox("Please enter Borrower Type", 0, "") ' Validation (Robert)
121         ElseIf MetroTextBox2.Text = "" Then
122             MsgBox("Please enter Borrower Name", 0, "") ' Validation (Robert)
123         ElseIf MetroTextBox3.Text = "" Then
124             MsgBox("Please enter Borrower Address", 0, "") ' Validation
125             (Robert)
126         ElseIf MetroTextBox4.Text = "" Then
127             MsgBox("Please enter Contact #", 0, "") ' Validation (Robert)
128         ElseIf MetroTextBox4.Text.Length > 11 Then
129             MsgBox("Minimum and Maximum of 11 digits", 0, "")
130         Else
131             Try
132                 If objcon.State = ConnectionState.Closed Then
133                     objcon.Open()
134                     com = New OleDb.OleDbCommand("INSERT INTO
135                     Borrowers values('" & MetroTextBox1.Text & "','" &
136                     MetroTextBox2.Text & "','" & MetroTextBox3.Text & "','" &
137                     MetroComboBox1.Text & "','" & MetroTextBox7.Text & "')" ,
138                     objcon)
139                     If com.ExecuteNonQuery() Then MsgBox("Saved Success!", 0,
140                     "")
141                     ListView2.Clear()
142                     Call ClearField()
143                     Call readData()
144                     objcon.Close()
145                     Call DisableThem()
146                     MetroTextBox1.Focus()
```



```
142         PictureBox4.Image = Nothing
143         Catch ex As Exception
144             MsgBox(ex.Message, 0, "")
145         End Try
146         PictureBox4.ResetText()
147     End If
148 End Sub
149
150 Private Sub MetroTextBox1_KeyPress(sender As Object, e As
151     KeyPressEventArgs) Handles MetroTextBox1.KeyPress
152     'condition for entering number only
153     ' If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
154     IsNumeric(e.KeyChar) Then
155         'MessageBox.Show("Please enter numbers only")
156         ' e.Handled = True
157         ' End If
158     End Sub
159
160 Private Sub MetroTextBox4_KeyPress(sender As Object, e As
161     KeyPressEventArgs) Handles MetroTextBox4.KeyPress
162     'condition for entering number only
163     If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
164     IsNumeric(e.KeyChar) Then
165         MessageBox.Show("Please enter numbers only")
166         e.Handled = True
167         End If
168
169 If MetroTextBox4.Text.Length > 10 Then
170     e.Handled = True
171     MessageBox.Show("Maximum 11 digits, Please try again")
172     MetroTextBox4.Text = ""
173     End If
174
175 End Sub
176
177 'condition for letters only (the best!!)
```



```
176 Private Sub MetroTextBox2_KeyPress(sender As Object, e As  
KeyPressEventArgs) Handles MetroTextBox2.KeyPress  
177  
178     If Not (Asc(e.KeyChar) = 8) Then  
179         If Not ((Asc(e.KeyChar) >= 97 And Asc(e.KeyChar) <=  
122) Or (Asc  
(e.KeyChar) >= 65 And Asc(e.KeyChar) <= 90 Or Asc(e.KeyChar) <=  
32) And Asc(e.KeyChar) >= 31) Then  
180             e.KeyChar = ChrW(0)  
181             e.Handled = True  
182         End If  
183     End If  
184  
185 End Sub  
186  
187  
188 Private Sub MetroButton8_Click(ByVal sender As System.Object, ByVal e  
As System.EventArgs) Handles MetroButton8.Click  
189     If MetroTextBox1.Text = "" Then  
190         MsgBox("Please enter the ID to be deleted!", 0, "")  
191     Else  
192         Try  
193             objcon.Open()  
194             com = New OleDb.OleDbCommand("delete from Borrowers  
where B_ID=" & MetroTextBox1.Text & "", objcon)  
195             If com.ExecuteNonQuery() Then  
196                 ListView2.Clear()  
197                 Call readData()  
198                 MsgBox("Deleted Success!", 0, "")  
199                 Call ClearField()  
200                 PictureBox4.Image = Nothing  
201                 MetroTextBox1.Focus()  
202             Else  
203                 MsgBox("ID Not Found!", 0, "")  
204             End If  
205  
206             objcon.Close()  
207             Catch ex As Exception  
208                 MsgBox(ex.Message, 0, "")
```



```
209         End Try
210     End If
211 End Sub
212
213 Private Sub MetroButton1_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles MetroButton1.Click
214     PictureBox4.Image = Nothing
215     Call EnableThem()
216     Call ClearField()
217     MetroTextBox1.Focus()
218 End Sub
219 Sub EnableThem()
220     MetroTextBox1.Enabled = True
221     MetroTextBox2.Enabled = True
222     MetroTextBox3.Enabled = True
223     MetroTextBox4.Enabled = True
224     MetroComboBox1.Enabled = True
225     MetroDateTime1.Visible = False
226
227 End Sub
228 Sub DisableThem()
229     'MetroTextBox1.Enabled = False
230     MetroTextBox2.Enabled = False
231     MetroTextBox3.Enabled = False
232     MetroTextBox4.Enabled = False
233     MetroComboBox1.Enabled = False
234     MetroDateTime1.Visible = False
235
236 End Sub
237
238
239
240 Sub readData()
241     ListView2.Columns.Add("RFID", 90, HorizontalAlignment.Center)
242     ListView2.Columns.Add("Borrower Name", 210,
HorizontalAlignment.Center)
243     ListView2.Columns.Add("Borrower Address", 130,
HorizontalAlignment.Center)
244     ListView2.Columns.Add("Contact #", 150, HorizontalAlignment.Center)
```



```
245     ListView2.Columns.Add("Borrower Type", 150,  
246     HorizontalAlignment.Center)  
247  
248     ListView2.View = View.Details  
249     Try  
250  
251         If (objcon.State = ConnectionState.Closed) Then objcon.Open()  
252         com = New OleDb.OleDbCommand("SELECT * FROM  
253         Borrowers", objcon) 253         dr = com.ExecuteReader  
254         While dr.Read()  
255             Call adddatatolistview(ListView2, dr(0), dr(1), dr(2), dr(3), dr(4),  
256             dr(5))  
257         End While  
258         dr.Close()  
259         objcon.Close()  
260         Catch  
261             'MsgBox("Please Refresh", MsgBoxStyle.Information, "")  
262         End Try  
263         End Sub  
264  
265     Public Sub adddatatolistview(ByVal lvw As ListView, ByVal B_ID As  
266     String,  
267     ByVal B_Name As String, ByVal B_Address As String, ByVal B_Cont As  
268     String, ByVal B_Type As String, ByVal B_Img As String)  
269     Dim lv As New ListViewItem  
270     lvw.Items.Add(lv)  
271     lv.Text = B_ID  
272     lv.SubItems.Add(B_Name)  
273     lv.SubItems.Add(B_Address)  
274     lv.SubItems.Add(B_Cont)  
275     lv.SubItems.Add(B_Type)  
276     lv.SubItems.Add(B_Img)  
277  
278     End Sub  
  
279     Sub ClearField()  
280     MetroTextBox1.Clear()
```



```
279 MetroTextBox2.Text = ""
280 MetroTextBox3.Text = ""
281 MetroTextBox4.Text = ""
282
283 MetroComboBox1.SelectedIndex = -1 'clearing combo box
284 MetroDateTime1.Value = DateTime.Now
285
286 End Sub
287 Sub LoadInto()
288
289     End Sub
290     Private Sub ListView2_SelectedIndexChanged(sender As Object,
291     e As EventArgs) Handles ListView2.SelectedIndexChanged
292     Dim i As Integer
293     For i = 0 To ListView2.Items.Count - 1
294     If ListView2.Items(i).Selected = True Then
295     MetroTextBox1.Text = ListView2.Items(i).SubItems(0).Text
296     MetroTextBox2.Text = ListView2.Items(i).SubItems(1).Text
297     MetroTextBox3.Text = ListView2.Items(i).SubItems(2).Text
298     MetroTextBox4.Text = ListView2.Items(i).SubItems(3).Text
299     MetroComboBox1.Text = ListView2.Items(i).SubItems(4).Text
300     PictureBox4.ImageLocation = ListView2.Items(i).SubItems
301     (5).Text
302
303     Exit For
304     End If
305     Next
306     ListView2.Focus()
307     ListView2.FullRowSelect = True
308     End Sub
309
310
311
312     Private Sub MetroTextBox1_TextChanged(ByVal sender
313     As System.Object, ByVal e As System.EventArgs)
314     Handles MetroTextBox1.TextChanged
315     Dim i As Integer
```



```
314         ListView2.SelectedItems.Clear()
315         MetroTextBox1.Focus()
316         Try
317         If Me.MetroTextBox1.Text = "" Then
318             MetroTextBox2.Text = ""
319             PictureBox4.Image = Nothing
320         Else
321             For i = 0 To ListView2.Items.Count - 1
322                 If MetroTextBox1.Text = ListView2.Items(i).SubItems
323                     (0).Text Then
324                         MetroTextBox2.Text = ListView2.Items(i).SubItems
325                         (1).Text
326                         PictureBox4.ImageLocation =
327                             MetroTextBox7.Text
328                         ListView2.Items(i).Selected = True
329                     Exit For
330                 End If
331             Next
332         End If
333         Catch
334     End Try
335 End Sub
336 Private Sub MetroButton3_Click(sender As Object, e As EventArgs)
337     Handles MetroButton3.Click
338     Try
339     Dim OFD As OpenFileDialog = New OpenFileDialog()
340
341     OFD.Filter = "Image File (*.jpg;*.bmp;*.gif)|*.jpg;*.bmp;*.gif"
342     If OFD.ShowDialog() = DialogResult.OK Then
343         PictureBox4.ImageLocation = OFD.FileName
344         MetroTextBox7.Text = OFD.FileName
345     End If
346
347     OFD = Nothing
348
349     Catch ex As Exception
350         MsgBox(ex.Message.ToString())
```



```
358         End Try
359     End Sub
360
361
362 End Class
363
```

A.3 AllBorrowed Class

1 Public Class AllBorrowed

System.EventArgs Handles MyBase.Load

```
4     Call readData()
5     End Sub
6     Sub readData()
7         ListView1.Clear()
8         ListView1.Columns.Add("", 60, HorizontalAlignment.Center)
9         ListView1.Columns.Add("", 60, HorizontalAlignment.Center)
10        ListView1.Columns.Add("", 310, HorizontalAlignment.Center)
11        ListView1.Columns.Add("", 90, HorizontalAlignment.Center)
12        ListView1.Columns.Add("", 90, HorizontalAlignment.Center)
13        ListView1.Columns.Add("", 190, HorizontalAlignment.Center)
14        ListView1.Columns.Add("", 90, HorizontalAlignment.Center)
15        ListView1.Columns.Add("", 90, HorizontalAlignment.Center)
16        ListView1.Columns.Add("", 100, HorizontalAlignment.Center)
17        ListView1.GridLines = True
18        ListView1.View = View.Details
19    Try
20
21        If (objcon.State = ConnectionState.Closed) Then objcon.Open()
22        com = New OleDb.OleDbCommand("SELECT * FROM Books
23        WHERE      ↗ status='Borrowed'", objcon)
24        dr = com.ExecuteReader
25        While dr.Read()
26            Call adddatatolistview(ListView1, dr(0), dr(1), dr(2), dr(3), ↗ dr(4),
27            dr(5), dr(6), dr(7))
28        End While
29        dr.Close()
30        objcon.Close()
31    Catch
32        'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
33    End Try
34 End Sub
```



```
31         End Try
32     End Sub
33     Public Sub adddatatolistview(ByVal lvw As ListView, ByVal
        BookID As String, ByVal GroupID As String, ByVal BookName
        As String, ByVal Publisher As
String, ByVal Author As String, ByVal PubYear As String, ByVal edi As
String, ByVal st As String)
34         Dim lv As New ListViewItem
35         lvw.Items.Add(lv)
36         lv.Text = BookID
37         lv.SubItems.Add(GroupID)
38         lv.SubItems.Add(BookName)
39         lv.SubItems.Add(Publisher)
40         lv.SubItems.Add(Author)
41         lv.SubItems.Add(PubYear)
42         lv.SubItems.Add(edi)
43         lv.SubItems.Add(st)
44     End Sub
45
46     Private Sub MetroButton1_Click(sender As Object, e As EventArgs)
        Handles MetroButton1.Click
47         LibraryMainForm.Show()
48         Me.Close()
49
50 End Sub
51 End Class
```

A.4 Archive Class

```
1 Public Class Archive
```

```
2
3     Private Sub MetroButton9_Click(sender As Object, e As EventArgs)
        Handles MetroButton9.Click
4         LibraryMainForm.Show()
5         Me.Close()
6
7 End Sub
8
```



```
9      Private Sub Archive_Load(sender As Object, e As EventArgs) Handles
      MyBase.Load
10      Call readData()
11      Call Name_Group()
12      'Call generateyear()
13
14  End Sub

15
16      Public Sub adddatatolistview(ByVal lvw As
      ListView,
17      ByVal BookID As String,
18      ByVal GroupID As String,
19      ByVal BookName As String,
20      ByVal Publisher As String,
21      ByVal Author As String,
22      ByVal PubYear As String,
23      ByVal edi As String,
24      ByVal cpy As String)
25      Dim lv As New ListViewItem
26      lvw.Items.Add(lv)
27      lv.Text = BookID
28      lv.SubItems.Add(GroupID)
29      lv.SubItems.Add(BookName)
30      lv.SubItems.Add(Publisher)
31      lv.SubItems.Add(Author)
32      lv.SubItems.Add(PubYear)
33      lv.SubItems.Add(edi)
34
35      lv.SubItems.Add(cpy)
36  End Sub

37
38  Sub readData()
39      MetroListView1.Clear()
40      MetroListView1.Columns.Add("Book Barcode", 160,
      HorizontalAlignment.Center)
41      MetroListView1.Columns.Add("Group ID", 80,
      HorizontalAlignment.Center)
42      MetroListView1.Columns.Add("Book Name", 290,
      HorizontalAlignment.Center)
```



```
43 MetroListView1.Columns.Add("Publisher", 90,
HorizontalAlignment.Center)
44 MetroListView1.Columns.Add("Author", 90, HorizontalAlignment.Center)
45 MetroListView1.Columns.Add("Publishing Year", 190,
HorizontalAlignment.Center)
46 MetroListView1.Columns.Add("Edition", 90, HorizontalAlignment.Center)
47 MetroListView1.Columns.Add("Copy", 105, HorizontalAlignment.Center)
48 MetroListView1.View = View.Details
49 Try
50
51     If (objcon.State = ConnectionState.Closed) Then objcon.Open()
52     com = New OleDb.OleDbCommand("SELECT * FROM
ArchivesBooks", objcon)
53     dr = com.ExecuteReader
54     While dr.Read()
55     Call adddatatolistview(MetroListView1, dr(0), dr(1), dr(2), dr
(3), dr(4), dr(5), dr(6), dr(9))
56     End While
57     dr.Close()
58     objcon.Close()
59     Catch
60     'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
61     End Try
62     End Sub
63
64 Private Sub MetroButton8_Click(sender As Object, e As EventArgs) Handles
MetroButton8.Click
65
66     Try
67     If objcon.State = ConnectionState.Closed Then objcon.Open() 68
69     If MessageBox.Show("Do you really want to delete?", "ARE YOU
SURE", MessageBoxButtons.YesNo) =
Windows.Forms.DialogResult.Yes Then
70
com3 = New OleDb.OleDbCommand("DELETE FROM
ArchivesBooks WHERE
Copy=@Copy AND BookID=@BID", objcon)
71
```



```
72         com3.Parameters.AddWithValue("@Copy", MetroTextBox6.Text)
73         com3.Parameters.AddWithValue("@BID", MetroTextBox1.Text)
74         com3.ExecuteNonQuery()
75         Call readData()
76
77     End If
78
79     Catch ex As Exception
80
81     End Try
82
83 End Sub
84
85
86     Sub Name_Group()
87     Try
88     If objcon.State = ConnectionState.Closed Then objcon.Open() 89
89     com = New OleDb.OleDbCommand("Select GroupName from
90     GroupName", objcon)
91     dr = com.ExecuteReader
92     While dr.Read
93     MetroComboBox1.Items.Add(dr.Item(0))
94     MetroComboBox1.Items.Add(dr.Item(7))
95
96     End While
97     dr.Close()
98     objcon.Close()
99     Catch ex As Exception
100
101     End Try
102     End Sub
103
104
105 Private Sub MetroListView1_SelectedIndexChanged(sender As Object, e As
EventArgs) Handles MetroListView1.SelectedIndexChanged
```



```
110         Dim i As Integer
111         For i = 0 To MetroListView1.Items.Count - 1
112             If MetroListView1.Items(i).Selected = True Then
113
114                 MetroTextBox1.Text =
115                     MetroListView1.Items(i).SubItems(0).Text
116                 MetroComboBox1.Text =
117                     MetroListView1.Items(i).SubItems(1).Text
118                 MetroTextBox2.Text =
119                     MetroListView1.Items(i).SubItems(2).Text
120                 MetroTextBox3.Text =
121                     MetroListView1.Items(i).SubItems(3).Text
122                 MetroTextBox4.Text =
123                     MetroListView1.Items(i).SubItems(4).Text
124                 MetroTextBox9.Text =
125                     MetroListView1.Items(i).SubItems(5).Text
126                 MetroTextBox7.Text =
127                     MetroListView1.Items(i).SubItems(6).Text
128                 MetroTextBox6.Text =
129                     MetroListView1.Items(i).SubItems(7).Text
130             Exit For
131         End If
132     Next
133     MetroListView1.Focus()
134     MetroListView1.FullRowSelect = True
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147
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149
150
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```



```
142      com.CommandText = "Select Max(Quantity) as maxid from Books Where
      BookID='" & MetroTextBox1.Text & "'"
143      maxid = com.ExecuteScalar
144
145      If maxid Is DBNull.Value Then
146          intID = 1
147      Else
148          Strid = CType(maxid, String)
149          intID = CType(Strid, String)
150          intID = intID + 1
151
152
153      End If
154
155      MetroTextBox10.Text = intID
156      objcon.Close()
157
158
159      Try
160      If objcon.State = ConnectionState.Closed Then objcon.Open() 161
162          If MessageBox.Show("Do you really want to restore?", "ARE
          YOU
          SURE", MessageBoxButtons.YesNo) =
          Windows.Forms.DialogResult.Yes Then
163              If intID = 1 Then
164                  com = New OleDb.OleDbCommand("INSERT INTO Books
                  VALUES("
                  & MetroTextBox1.Text & "','" & MetroComboBox1.Text &
                  "','"
                  & MetroTextBox2.Text & "','" & MetroTextBox3.Text & "','"
                  & MetroTextBox4.Text & "','" & MetroTextBox9.Text & "','"
                  & MetroTextBox7.Text & "','" & MetroTextBox10.Text & "','"
                  & MetroTextBox5.Text & "')", objcon)
165
166      com3 = New OleDb.OleDbCommand("DELETE FROM ArchivesBooks
          WHERE Copy=@Copy AND BookID=@BID", objcon)
```



```
167
168         com3.Parameters.AddWithValue("@Copy",
169         MetroTextBox6.Text)
170         com3.Parameters.AddWithValue("@BID",
171         MetroTextBox1.Text)
172         com3.ExecuteNonQuery()
173         com.ExecuteNonQuery()
174         MsgBox("Restored successfully 1", 0, "SUCCESS")
175         Call readData()
176
177         ElseIf intID > 1 Then
178         com2 = New OleDb.OleDbCommand("UPDATE Books SET
179         Quantity='" & MetroTextBox10.Text & "' WHERE
180         BookID='" & MetroTextBox1.Text & "'", objcon)
181
182         com3.Parameters.AddWithValue("@Copy",
183         MetroTextBox6.Text)
184         com3.Parameters.AddWithValue("@BID",
185         MetroTextBox1.Text)
186         com3.ExecuteNonQuery()
187
188         com2.ExecuteNonQuery()
189         MsgBox("Restored successfully 2", 0, "SUCCESS")
190         Call readData()
191
192     End If
193
194 End If
195
196
197 End If
198
199
200
201 Catch ex As Exception
202
203
204 End Try
205
206
```



```
207 End Sub
208
211 Private Sub MetroTextBox6_TextChanged(sender As Object, e As
    EventArgs) Handles MetroTextBox6.TextChanged
212 '////////////////////////////////////
213 If objcon.State = ConnectionState.Closed Then objcon.Open() 214
215 com = New OleDb.OleDbCommand("SELECT * FROM Books", objcon)
216 com.Connection = objcon
217 Dim maxid As Object
218 Dim Strid As String
219 Dim intID As Integer
220
221 com.CommandText = "Select Max(Quantity) as maxid from Books Where
    BookID='" & MetroTextBox1.Text & "'"
222 maxid = com.ExecuteScalar
223
224 If maxid Is DBNull.Value Then
225     intID = 1
226 Else
227     Strid = CType(maxid, String)
228     intID = CType(Strid, Integer)
229     intID = intID + 1
230
231
232 End If
233
234 MetroTextBox10.Text = intID
235 objcon.Close()
236
237 End Sub
238 End Class
```

A.5 AttendanceMainForm

```
1
2
3 Imports System
4 Imports System.IO.Ports
5
6 Public Class AttendanceMainForm
```



```
7      Dim receivedData As String = ""
8
9      Function ReceiveSerialData() As String
10         receivedData = ""
11         Dim Reader As String
12         Try
13
14             Reader = SerialPort2.ReadExisting()
15             If Reader Is Nothing Then
16                 Return "Nothing" & vbCrLf
17             Else
18                 Return Reader
19             End If
20             Catch ex As TimeoutException
21                 Return "Error"
22             End Try
23
24     End Function
25     Function ArduinoConnect() As String
26
27
28         For Each sp As String In My.Computer.Ports.SerialPortNames
29             SerialPort2.Close()
30             SerialPort2.PortName = sp
31             SerialPort2.BaudRate = 9600
32             SerialPort2.DataBits = 8
33             SerialPort2.Parity = Parity.None
34             SerialPort2.StopBits = StopBits.One
35             SerialPort2.Handshake = Handshake.None
36             SerialPort2.Encoding = System.Text.Encoding.Default 'very
important!
37             SerialPort2.ReadTimeout = 10000
38             SerialPort2.Open()
39             Timer2.Enabled = True
40         Next
41     End Function
42
43     Private Sub AttendanceMainForm_FormClosing(sender As Object, e As
FormClosingEventArgs) Handles Me.FormClosing
```



```
46         LoginForm2.Show()
47         LoginForm2.UserNameTextBox2M.Focus()
51     End Sub
55     Private Sub AttendanceMainForm_Load(ByVal sender As System.Object,
56         ByVal e As System.EventArgs) Handles MyBase.Load
57         Call readData1()
58         MetroTextBox10.Focus()
59         ArduinoConnect()
60         Timer1.Enabled = True
61
62     End Sub
63
64     Private Sub Timer2_Tick(sender As Object, e As EventArgs) Handles
65         Timer2.Tick
66         receivedData = ReceiveSerialData()
67         MetroTextBox10.Text &= receivedData
68
69
70
71
72
73
74
75
76
77
78     End Sub
79
80
81
82
83     Private Sub MetroButton2_Click(sender As Object, e As EventArgs) Handles
84         MetroButton2.Click
85         'MainForm.Show()
86         'Me.Close()
87         LoginForm2.Show()
88         LoginForm2.UserNameTextBox2M.Focus()
139         Public Sub adddatatolistview1(ByVal lvw1 As
140             ListView,
141             ByVal B_ID As String,
142             ByVal B_Name As String,
143             ByVal B_Address As String,
144             ByVal B_Type As String,
145             ByVal B_ContactNumber As String,
146             ByVal B_Img As String)
```



```
147         Dim lv1 As New ListViewItem
148         lvw1.Items.Add(lv1)
149         lv1.Text = B_ID
150         lv1.SubItems.Add(B_Name)
151         lv1.SubItems.Add(B_Address)
152         lv1.SubItems.Add(B_Type)
153         lv1.SubItems.Add(B_ContactNumber)
154         lv1.SubItems.Add(B_Img)
155
156
157
158     End Sub
159
160     Sub readData1()
161         ListView1.Clear()
162         ListView1.Columns.Add("", 60, HorizontalAlignment.Center)
163         ListView1.Columns.Add("", 60, HorizontalAlignment.Center)
164         ListView1.Columns.Add("", 310, HorizontalAlignment.Center)
165         ListView1.Columns.Add("", 90, HorizontalAlignment.Center)
166         ListView1.Columns.Add("", 90, HorizontalAlignment.Center)
167         ListView1.View = View.Details
168         Try
169             If (objcon.State = ConnectionState.Closed) Then objcon.Open()
170             com = New OleDb.OleDbCommand("SELECT * FROM
171             Borrowers ", objcon)
172             dr = com.ExecuteReader
173             While dr.Read()
174                 Call adddatatolistview1(ListView1, dr(0), dr(1), dr(2), dr(3),
175                 dr(4), dr(5))
176             End While
177             dr.Close()
178             objcon.Close()
179             Catch
180                 'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
181             End Try
182
183
184
185     End Sub
186
```



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```
190
191 Private Sub ListView1_SelectedIndexChanged_1(sender As Object, e As
    EventArgs) Handles ListView1.SelectedIndexChanged
192
193     Dim i As Integer
194     For i = 0 To ListView1.Items.Count - 1
195         If ListView1.Items(i).Selected = True Then
196
197             Exit For
198         End If
199     Next
200     ListView1.Focus()
201     ListView1.FullRowSelect = True
202 End Sub
203
204 Private Sub MetroTextBox10_KeyPress(sender As Object, e As
    KeyPressEventArgs) Handles MetroTextBox10.KeyPress
205     If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
        IsNumeric(e.KeyChar) Then
206         MessageBox.Show("Error")
207         e.Handled = True
208     End If
209 End Sub
210
211
212 Private Sub MetroTextBox10_TextChanged(sender As Object, e As
    EventArgs) Handles MetroTextBox10.TextChanged
213     Dim i As Integer
214     ListView1.SelectedItems.Clear()
215     MetroTextBox10.Focus()
216
217 Try
221     If Me.MetroTextBox10.Text = "" Then
222         ' Call CellRestart()
223         MetroTextBox1.Text = ""
224         MetroTextBox7.Text = ""
225         MetroTextBox2.Text = ""
226         PictureBox4.Image = Nothing
227     Else
```



```
228
229
230         For i = 0 To ListView1.Items.Count - 1
231             If MetroTextBox10.Text = ListView1.Items(i).SubItems
232                 (0).Text Then
233                     MetroTextBox1.Text = ListView1.Items(i).SubItems
234                     (1).Text
235                     MetroTextBox7.Text = ListView1.Items(i).SubItems
236                     (2).Text
237                     MetroTextBox2.Text = ListView1.Items(i).SubItems
238                     (4).Text
239                     PictureBox4.ImageLocation = ListView1.Items
240                     (i).SubItems(5).Text
241                     MetroTextBox10.Focus()
242                     'MetroLabel6.Hide()
243                     Exit For
244             Else
245                 'MetroLabel6.Show()
246                 End If
247         Next
248     End If
249     ' Call locateCell()
250
251     Catch
252
253     End Try
254
255 End Sub
262
263 Private Sub MetroTextBox2_TextChanged(sender As Object, e As EventArgs)
264     Handles MetroTextBox2.TextChanged
265
266     If MetroTextBox2.Text <> "" Then
```



```
267
268
269     If objcon.State = ConnectionState.Closed Then objcon.Open()
270     com2 = New OleDb.OleDbCommand("SELECT * FROM
QAttendance WHERE
    A_Name='" & MetroTextBox1.Text & "' AND A_Date='" &
    MetroLabel7.Text & "'", objcon)
271     dr = com2.ExecuteReader
272
273     'If (dr("A_Action").ToString()) = "Login" Then
274     ' MsgBox("Detected Login")
275     'Else
276     ' MsgBox("Detected Logout")
277     'End If
278
279     If (dr.HasRows) Then
280     If objcon.State = ConnectionState.Closed Then objcon.Open() 281
282         com4 = New OleDb.OleDbCommand("DELETE *
FROM QAttendance WHERE A_Name='" &
MetroTextBox1.Text & "'", objcon)
283         com4.ExecuteNonQuery()
284         '////////////////////////////////////
285         Try
286         If objcon.State = ConnectionState.Closed Then
objcon.Open ()
287         com = New OleDb.OleDbCommand("INSERT INTO
Attendance values('" & MetroTextBox1.Text & "','"
&
MetroTextBox7.Text & "','" & MetroTextBox2.Text & "','" &
MetroLabel6.Text & "','Logout,'" & MetroLabel7.Text &
"'")", objcon)
288
289         If com.ExecuteNonQuery() Then
290         objcon.Close()
291         MetroTextBox1.Clear()
292         MetroTextBox7.Clear()
293         MetroTextBox2.Clear()
294
295         MetroTextBox10.Clear()
```



```
296         PictureBox4.Image = Nothing
297         'MsgBox("Logout")
298         End If
299
300
301     Catch ex As Exception
302         MsgBox(ex.Message, 0, "adaw")
303     End Try
304     PictureBox4.ResetText()
305     'SerialPort2.Close()
306     'AttendanceMainForm2.Show()
307     'Me.Hide()
308     '////////////////////////////////////
309     Else
310
311         com3 = New OleDb.OleDbCommand("INSERT INTO
312         QAttendance values ('" & MetroTextBox1.Text & "','" & MetroTextBox7.Text &
313         "','" & MetroTextBox2.Text & "','" & MetroLabel6.Text &
314         "','" & MetroTextBox3.Text & "','" & MetroLabel7.Text & "')" ,
315         objcon)
316         com3.ExecuteNonQuery()
317         '////////////////////////////////////
318         Try
319             If objcon.State = ConnectionState.Closed Then
320                 objcon.Open ()
321                 com = New OleDb.OleDbCommand("INSERT INTO
322                 Attendance values('" & MetroTextBox1.Text & "','" & MetroTextBox7.Text & "','" & MetroTextBox2.Text & "','" & MetroLabel6.Text & "','" & MetroLabel7.Text & "')" ,
323                 objcon)
324                 If com.ExecuteNonQuery() Then
325                     objcon.Close()
326                     MetroTextBox1.Clear()
327                     MetroTextBox7.Clear()
328                     MetroTextBox2.Clear()
329
330                     MetroTextBox10.Clear()
331                     PictureBox4.Image = Nothing
```



```
325         ' MsgBox("login")
326     End If
327     Catch ex As Exception
328         MsgBox(ex.Message, 0, "adaw")
329     End Try
330     PictureBox4.ResetText()
331     ' SerialPort2.Close()
332     ' AttendanceMainForm2.Show()
333     ' Me.Hide()
334     '////////////////////
335     End If
342
343 End If
344 End Sub
345
346 Private Sub MetroButton3_Click(sender As Object, e As EventArgs)
347     Handles MetroButton3.Click
348     Try
349         If objcon.State = ConnectionState.Closed Then objcon.Open()
350         com = New OleDb.OleDbCommand("DELETE * FROM Attendance",
351         objcon)
352         If com.ExecuteNonQuery() Then
353             MsgBox("Submitted!", 0, "")
354
355         objcon.Close()
356
357         Catch ex As Exception
358             MsgBox(ex.Message, 0, "")
359         End Try
360     End Sub
361
362 Private Sub Timer1_Tick(sender As Object, e As EventArgs) Handles Timer1.Tick
363     MetroLabel6.Text = Date.Now.ToString("hh:mm:ss tt")
364     MetroLabel7.Text = Date.Now.ToString("MM/dd/yyyy")
365 End Sub
End Class
```

A.6 Book Detail



```
1 Public Class BookDetail
2
3 Private Sub BookDetail_Load(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles MyBase.Load
4
5     ' MetroTabControl2.Visible = False
6
7     Call readData() 8 End Sub
9
10 Public Sub adddatatolistview(ByVal lvw As ListView, ByVal BookID As String,
    ByVal GroupID As String, ByVal BookName As String, ByVal publisher As
    String, ByVal author As String, ByVal pubyear As String, ByVal edi As
    String, ByVal qty As String) 11 Dim lv As New ListViewItem
12     lvw.Items.Add(lv)
13     lv.Text = BookID
14     lv.SubItems.Add(GroupID)
15     lv.SubItems.Add(BookName)
16     lv.SubItems.Add(publisher)
17     lv.SubItems.Add(author)
18     lv.SubItems.Add(pubyear)
19     lv.SubItems.Add(edi)
20     lv.SubItems.Add(qty)
21 End Sub
25 Sub readData()
26     MetroListView2.Clear()
27     MetroListView2.Columns.Add("Book Barcode", 160,
    HorizontalAlignment.Left)
28     MetroListView2.Columns.Add("Group ID", 160,
    HorizontalAlignment.Center)
29     MetroListView2.Columns.Add("Book Name", 210,
    HorizontalAlignment.Center)
30     MetroListView2.Columns.Add("Publisher", 90,
    HorizontalAlignment.Center)
31     MetroListView2.Columns.Add("Author", 90, HorizontalAlignment.Center)
32     MetroListView2.Columns.Add("Publishing Year", 190,
    HorizontalAlignment.Center)
33     MetroListView2.Columns.Add("Edition", 90, HorizontalAlignment.Center)
34     MetroListView2.Columns.Add("Quantity", 90,
    HorizontalAlignment.Center)
```



```
35
36
37 MetroListView2.View = View.Details
38 Try
39
40     If (objcon.State = ConnectionState.Closed) Then objcon.Open()
41     com = New OleDb.OleDbCommand("SELECT * FROM Books", objcon)
42     dr = com.ExecuteReader
43     While dr.Read()
44         Call adddatatolistview(MetroListView2, dr(0), dr(1), dr(2), dr
            (3), dr(4), dr(5), dr(6), dr(7))
45     End While
46     dr.Close()
47     objcon.Close()
48     Catch
49         'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
50 End Try 51 End Sub
59 Private Sub Button6_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) 60 Try 61 Dim i As Integer
62     For i = 0 To MetroListView2.Items.Count - 1
63     If MetroListView2.Items(i).Selected = True Then
64     metrotextbox1.Text = MetroListView2.Items(i + 1).SubItems
        (0).Text
65     Exit For 66 End If 67 Next
68     MetroListView2.Focus()
69     MetroListView2.FullRowSelect = True
70     Catch ex As Exception
71
72 End Try 73 End Sub
74
75 Private Sub Button5_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) 76 Try 77 Dim i As Integer
78     For i = 0 To MetroListView2.Items.Count - 1
79     If MetroListView2.Items(i).Selected = True Then
80     metrotextbox1.Text = MetroListView2.Items(i - 1).SubItems
        (0).Text
```



```
81      Exit For 82      End If 83      Next
84      MetroListView2.Focus()
85      MetroListView2.FullRowSelect = True
86      Catch ex As Exception
87
88  End Try 89  End Sub
93 Private Sub Button1_Click(sender As Object, e As EventArgs)
94     LibraryMainForm.Show()
95     Me.Hide()
96 End Sub
97
98     Private Sub MetroButton2_Click(sender As Object, e As EventArgs)
99         Handles MetroButton2.Click
100        LibraryMainForm.Show()
101        Me.Close()
102 End Sub
103
104 End Class
```

A.7 Borrower Detail

```
1 Public Class BookDetail
2
3 Private Sub BookDetail_Load(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles MyBase.Load
4
5     ' MetroTabControl2.Visible = False
6
7     Call readData() 8 End Sub
9
10 Public Sub adddatatolistview(ByVal lvw As ListView, ByVal BookID As String,
    ByVal GroupID As String, ByVal BookName As String, ByVal publisher As
    String, ByVal author As String, ByVal pubyear As String, ByVal edi As
    String, ByVal qty As String) 11 Dim lv As New ListViewItem
21     lvw.Items.Add(lv)
22     lv.Text = BookID
23     lv.SubItems.Add(GroupID)
24     lv.SubItems.Add(BookName)
25     lv.SubItems.Add(publisher)
```



```
26         lv.SubItems.Add(author)
27         lv.SubItems.Add(pubyear)
28         lv.SubItems.Add(edi)
29         lv.SubItems.Add(qty)
21     End Sub
22
23
24
35     Sub readData()
36         MetroListView2.Clear()
37         MetroListView2.Columns.Add("Book Barcode", 160,
HorizontalAlignment.Left)
38         MetroListView2.Columns.Add("Group ID", 160,
HorizontalAlignment.Center)
39         MetroListView2.Columns.Add("Book Name", 210,
HorizontalAlignment.Center)
40         MetroListView2.Columns.Add("Publisher", 90,
HorizontalAlignment.Center)
41         MetroListView2.Columns.Add("Author", 90, HorizontalAlignment.Center)
42         MetroListView2.Columns.Add("Publishing Year", 190,
HorizontalAlignment.Center)
43         MetroListView2.Columns.Add("Edition", 90,
HorizontalAlignment.Center)
44         MetroListView2.Columns.Add("Quantity", 90,
HorizontalAlignment.Center)
35
36
37     MetroListView2.View = View.Details
38     Try
39
43         If (objcon.State = ConnectionState.Closed) Then objcon.Open()
44         com = New OleDb.OleDbCommand("SELECT * FROM Books",
objcon)
45         dr = com.ExecuteReader
43     While dr.Read()
44         Call adddatatolistview(MetroListView2, dr(0), dr(1), dr(2), dr
(3), dr(4), dr(5), dr(6), dr(7))
```



```
45      End While
50      dr.Close()
51      objcon.Close()
52      Catch
53      'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
50 End Try 51 End Sub
52
53
54
55
56
57
58
59 Private Sub Button6_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) 60      Try 61      Dim i As Integer
65          For i = 0 To MetroListView2.Items.Count - 1
66          If MetroListView2.Items(i).Selected = True Then
67          metrotextbox1.Text = MetroListView2.Items(i + 1).SubItems
            (0).Text
65      Exit For 66      End If 67      Next
71      MetroListView2.Focus()
72      MetroListView2.FullRowSelect = True
73      Catch ex As Exception
71
72 End Try 73 End Sub
74
75 Private Sub Button5_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) 76      Try 77      Dim i As Integer
81          For i = 0 To MetroListView2.Items.Count - 1
82          If MetroListView2.Items(i).Selected = True Then
83          metrotextbox1.Text = MetroListView2.Items(i - 1).SubItems
            (0).Text
81      Exit For 82      End If 83      Next
87      MetroListView2.Focus()
88      MetroListView2.FullRowSelect = True
89      Catch ex As Exception
87
```



```
88     End Try 89     End Sub
90
91
92
93 Private Sub Button1_Click(sender As Object, e As EventArgs)
94     LibraryMainForm.Show()
95     Me.Hide()
96 End Sub
97
101 Private Sub MetroButton2_Click(sender As Object, e As EventArgs)
102     Handles MetroButton2.Click
103     LibraryMainForm.Show()
104     Me.Close()
105 End Sub
106
107
108 End Class
```

A.8 GroupID

```
1 Public Class GroupID
2     Public NameFrm, NameTo As String
3     Private Sub MetroButton3_Click(ByVal sender As System.Object, ByVal e
4         As System.EventArgs) Handles MetroButton3.Click
5         LibraryMainForm.Show()
6         Me.Close()
7     End Sub
8
9     Private Sub MetroButton2_Click(ByVal sender As System.Object, ByVal e
10         As System.EventArgs) Handles MetroButton2.Click
11         MetroTextBox1.Clear()
12         MetroTextBox2.Clear()
13     End Sub
14     Sub disablethem()
15         'MetroTextBox1.Enabled = False
16         MetroTextBox1.Enabled = False
17         MetroTextBox2.Enabled = False
18     End Sub
19     Sub enablethem()
```



```
19      'MetroTextBox1.Enabled = False
20      MetroTextBox1.Enabled = True
21      MetroTextBox2.Enabled = True
22
23  End Sub

24
25
26  Private Sub MetroButton1_Click(ByVal sender As System.Object, ByVal e As
      System.EventArgs) Handles MetroButton1.Click
27
28      If objcon.State = ConnectionState.Closed Then objcon.Open()
29
30      com = New OleDb.OleDbCommand("SELECT * FROM GroupName",
      objcon)
31      com.Connection = objcon
32      Dim maxid As Object
33      Dim Strid As String
34      Dim intID As Integer
35
36      com.CommandText = "Select Max(GroupID) as maxid from
      GroupName"
37      maxid = com.ExecuteScalar
38
39      If maxid Is DBNull.Value Then
40          intID = 1
41      Else
42          Strid = CType(maxid, String)
43          intID = CType(Strid, Integer)
44          intID = intID + 1
45
46
47  End If
48
49      MetroTextBox1.Text = intID
50      objcon.Close()
51
52
53      If MetroTextBox2.Text = "" Then
```



```
54         MsgBox("Please enter a Group Name!", 0, "")
55     Else
56     Try
57     If objcon.State = ConnectionState.Closed Then objcon.Open()
58     com = New OleDb.OleDbCommand("INSERT INTO GroupName
VALUES(" & MetroTextBox1.Text & "," &
MetroTextBox2.Text & ")", objcon)
59     com.ExecuteNonQuery()
60     MetroListView2.Clear()
61     Call readData()
62     MsgBox("Successfully Saved", 0, "")
63     objcon.Close()
64     MetroButton7.Show()
65     MetroListView2.Enabled = True
66     MetroButton6.Enabled = True
67
68     Catch ex As Exception
69     MsgBox(ex.Message, 0, "")
70     End Try
71     End If
72     End Sub
73
74     Private Sub GroupID_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load
75     Call readData()
76     End Sub
77     Sub readData()
78     MetroListView2.Columns.Add("GROUP ID", 100,
HorizontalAlignment.Center)
79     MetroListView2.Columns.Add("GROUP NAME", 300,
HorizontalAlignment.Center)
80     MetroListView2.View = View.Details
81     Try
82
83     If (objcon.State = ConnectionState.Closed) Then objcon.Open()
84     com = New OleDb.OleDbCommand("SELECT * FROM
GroupName", objcon)
85     dr = com.ExecuteReader86     While dr.Read()
87     Call disablethem()
```



```
88         Call adddatatolistview(MetroListView2, dr(0), dr(1))
89     End While
90     dr.Close()
91     objcon.Close()
92     Catch
93     'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
94 End Try
95 End Sub
96 Public Sub adddatatolistview(ByVal lvw As ListView, ByVal
GroupID As String, ByVal GroupName As String)
97     Dim lv As New ListViewItem
98     lvw.Items.Add(lv)
99     lv.Text = GroupID
100    lv.SubItems.Add(GroupName)
101 End Sub
102 Sub Sentence()
103     Dim a, b As Integer
104     a = NameFrm.Length105     NameTo = ""
106     For b = 0 To a - 1
107         If b = 0 Then
108             If Char.IsLower(NameFrm(0)) Then
109                 NameTo = Char.ToUpper(NameFrm(0))
110             Else
111                 NameTo = NameFrm(0)
112             End If
113         Else
114             If NameFrm(b - 1) = " " Then
115                 NameTo = NameTo + Char.ToUpper(NameFrm(b))
116             Else
117                 NameTo = NameTo + NameFrm(b)
118             End If
119         End If
120     Next
121 End Sub
122
123 Private Sub MetroTextbox2_LostFocus(ByVal sender As Object, ByVal e
As System.EventArgs) Handles MetroTextBox2.LostFocus
124     NameFrm = MetroTextbox2.Text
125     Call Sentence()
```



```
126         MetroTextbox2.Text = NameTo
127     End Sub
128
129 Private Sub MetroTextbox2_TextChanged(ByVal sender As System.Object, ByVal
    e As System.EventArgs) Handles MetroTextBox2.TextChanged
130
131 End Sub
132
133 Private Sub MetroTextbox1_TextChanged(ByVal sender
    As System.Object, ByVal e As System.EventArgs)
    Handles MetroTextBox1.TextChanged
134     Dim i As Integer
135     MetroListView2.SelectedItems.Clear()
136     MetroTextbox1.Focus()
137     Try
138     If Me.MetroTextbox1.Text = "" Then
139         MetroTextbox2.Text = ""
140     Else
141     For i = 0 To MetroListView2.Items.Count - 1
142         If MetroTextbox1.Text =
            MetroListView2.Items(i).SubItems
            (0).Text Then
143             MetroTextbox2.Text =
                MetroListView2.Items(i).SubItems
                (1).Text
144             MetroListView2.Items(i).Selected = True
145         Exit For
146     End If
147     Next
148     End If
149     Catch
150
151 End Try
152 End Sub
153
154 Private Sub MetroButton4_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles MetroButton4.Click
```



```
155         If MetroTextBox1.Text = "" Then
156             MsgBox("Please enter Group ID!", 0, "")
157         Else
158
159             Try
160                 If objcon.State = ConnectionState.Closed Then objcon.Open()
161                 com = New OleDb.OleDbCommand("DELETE FROM
Group Name WHERE GroupID=" & MetroTextBox1.Text & "",
objcon)
162                 com.ExecuteNonQuery()
163                 MetroListView2.Clear()
164                 Call readData()
165                 MsgBox("Deleted", 0, "")
166                 objcon.Close()
167             Catch ex As Exception
168                 MsgBox("Group ID number not found!", 0, "")
169             End Try
170         End If
171     End Sub
172
173     Private Sub MetroButton5_Click(ByVal sender As
System.Object, ByVal e As System.EventArgs) Handles
MetroButton5.Click
174         If MetroTextBox1.Text = "" Then
175             MsgBox("Please enter a Group ID!", 0, "")
176         Else
177             Try
178                 If objcon.State = ConnectionState.Closed Then objcon.Open()
179                 com = New OleDb.OleDbCommand("UPDATE Group Name SET
Group Name=" & MetroTextBox2.Text & " WHERE
GroupID=" & MetroTextBox1.Text & "", objcon)
180                 com.ExecuteNonQuery()
181                 MetroListView2.Clear()
182                 Call readData()
183                 MsgBox("Update successfully", 0, "")
184                 objcon.Close()
185                 MetroButton6.Show()
186                 MetroButton1.Enabled = True
187                 MetroButton7.Enabled = True
```



```
188
189
190         Catch ex As Exception
191             MsgBox("Cannot Update", 0, "")
192         End Try
193         End If
194         End Sub
195
196
197     Private Sub MetroButton3_Click_1(sender As Object, e As EventArgs)
198         Me.Hide()
199         LibraryMainForm.Show()
200     End Sub
201
202     Private Sub MetroListView2_SelectedIndexChanged_1(sender As
203         Object, e As EventArgs) Handles
204             MetroListView2.SelectedIndexChanged
205             Dim i As Integer
206             For i = 0 To MetroListView2.Items.Count - 1
207                 If MetroListView2.Items(i).Selected = True Then
208                     MetroTextBox1.Text =
209                         MetroListView2.Items(i).SubItems(0).Text
210                 Exit For
211             End If
212             Next
213             MetroListView2.Focus()
214             MetroListView2.FullRowSelect = True
215             End Sub
216
217     Private Sub MetroButton6_Click(sender As Object, e As EventArgs)
218         Handles MetroButton6.Click
219         MetroButton6.Hide()
220         MetroButton5.Show()
221         MetroButton1.Enabled = False
222         MetroButton7.Enabled = False
223         enablethem()
224     End Sub
225
```



```
223 Private Sub MetroButton7_Click(sender As Object, e As EventArgs)
    Handles MetroButton7.Click
224 MetroButton7.Hide()
225 MetroButton5.Show()
226 enablethem()
227 MetroTextBox1.Text = ""
228 MetroTextBox2.Text = ""
229 MetroListView2.Enabled = False
230 MetroButton6.Enabled = False
231 End Sub
232 End Class
```

A.9 Issue Book

```
1
2
3 Imports System
4 Imports System.IO.Ports
5
6 Public Class IssueBook
7
8 Dim receivedData As String = ""
9
10 Function ReceiveSerialData() As String
11 receivedData = ""
12 Dim Reader As String
13 Try
14
15 Reader = SerialPort2.ReadExisting()
16 If Reader Is Nothing Then
17 Return "Nothing" & vbCrLf
18 Else
19 Return Reader
20 End If
21 Catch ex As TimeoutException
22 Return "Error"
23 End Try
24
25
26 End Function
```



```
27
28 '////////////////////
29
30     Function ArduinoConnect() As String
31     Timer2.Enabled = False
32
33         For Each sp As String In My.Computer.Ports.SerialPortNames
34             SerialPort2.Close()
35             SerialPort2.PortName = sp
36             SerialPort2.BaudRate = 9600
37             SerialPort2.DataBits = 8
38             SerialPort2.Parity = Parity.None
39             SerialPort2.StopBits = StopBits.One
40             SerialPort2.Handshake = Handshake.None
41             SerialPort2.Encoding = System.Text.Encoding.Default 'very
42             important!
43             SerialPort2.ReadTimeout = 10000
44             SerialPort2.Open()
45             Timer2.Enabled = True
46         Next
47     End Function
48
49     Private Sub IssueBook_Load(ByVal sender As System.Object, ByVal e As
50     System.EventArgs) Handles MyBase.Load
51     MetroTextBox8.Select()
52     MetroDateTime1.MinDate = DateTime.Today
53     MetroDateTime1.Enabled = False
54     MetroDateTime2.MinDate = DateTime.Today
55     Call readData()
56     MetroDateTime2.Value = MetroDateTime2.Value.AddDays(3)
57     ArduinoConnect()
58     End Sub
59
60     Private Sub Timer2_Tick(sender As Object, e As EventArgs) Handles
61     Timer2.Tick
62     receivedData = ReceiveSerialData()
63     MetroTextBox9.Text &= receivedData
```



```
63         End Sub
64
65
66
67     Private Sub metrobutton4_Click(ByVal sender As System.Object, ByVal e
        As System.EventArgs)
68         Me.Close()
69         LibraryMainForm.Show()
70     End Sub
71
72     Private Sub MetroTextBox8_KeyPress(sender As Object, e As
        KeyPressEventArgs) Handles MetroTextBox8.KeyPress
73         'condition for entering number only
74         If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8
        AndAlso Not IsNumeric(e.KeyChar) Then
75             MessageBox.Show("Please enter numbers only")
76             e.Handled = True
77         End If
78     End Sub
79
80
81
82
83     Sub readData()
84         MetroListView1.Clear()
85         MetroListView1.Columns.Add("Book Barcode", 160,
        HorizontalAlignment.Center)
86         MetroListView1.Columns.Add("Group ID", 60,
        HorizontalAlignment.Center)
87         MetroListView1.Columns.Add("Book Name", 310,
        HorizontalAlignment.Center)
88         MetroListView1.Columns.Add("Publisher", 90,
        HorizontalAlignment.Center)
89         MetroListView1.Columns.Add("Author", 90, HorizontalAlignment.Center)
90         MetroListView1.Columns.Add("Publishing Year", 190,
        HorizontalAlignment.Center)
91         MetroListView1.Columns.Add("Edition", 90, HorizontalAlignment.Center)
```



```
92 MetroListView1.Columns.Add("Quantity", 90,  
HorizontalAlignment.Center) 93 MetroListView1.View =  
View.Details  
94  
95 Try  
96  
97 If (objcon.State = ConnectionState.Closed) Then objcon.Open()  
98 com = New OleDb.OleDbCommand("SELECT * FROM Books",  
objcon)  
99 dr = com.ExecuteReader  
100 While dr.Read()  
101 Call adddatatolistview(MetroListView1, dr(0), dr(1), dr(2), dr  
(3), dr(4), dr(5), dr(6), dr(7))  
102 End While  
103 dr.Close()  
104 objcon.Close()  
105 Catch  
106 MsgBox("Please Refresh", MsgBoxStyle.Information, "")  
107 End Try  
108 End Sub  
109 Public Sub adddatatolistview(ByVal lvw As ListView, ByVal  
BookID As String, ByVal GroupID As String, ByVal BookName  
As String, ByVal  
Publisher As String, ByVal Author As String, ByVal PubYear As String,  
ByVal edi As String, ByVal qty As String)  
110 Dim lv As New ListViewItem  
111 lvw.Items.Add(lv)  
112 lv.Text = BookID  
113 lv.SubItems.Add(GroupID)  
114 lv.SubItems.Add(BookName)  
115 lv.SubItems.Add(Publisher)  
116 lv.SubItems.Add(Author)  
117 lv.SubItems.Add(PubYear)  
118 lv.SubItems.Add(edi)  
119  
120 lv.SubItems.Add(qty)  
121 End Sub  
122  
123
```



```
124
125     Private Sub MetroTextBox8_SelectedIndexChanged(ByVal sender As
126         System.Object, ByVal e As System.EventArgs)
127         Dim i As Integer
128         MetroListView1.SelectedItems.Clear()
129
130         Try
131             If Me.MetroTextBox8.Text = "" Then
132                 MetroTextBox6.Text = ""
133             Else
134                 For i = 0 To MetroListView1.Items.Count - 1
135                     If MetroTextBox8.Text =
136                         MetroListView1.Items(i).SubItems
137                         (0).Text Then
138                             MetroTextBox6.Text =
139                             MetroListView1.Items(i).SubItems (1).Text
140                             MetroTextBox2.Text =
141                             MetroListView1.Items(i).SubItems (2).Text
142                             MetroTextBox3.Text =
143                             MetroListView1.Items(i).SubItems (3).Text
144                             MetroTextBox4.Text =
145                             MetroListView1.Items(i).SubItems (4).Text
146                             MetroTextBox7.Text =
147                             MetroListView1.Items(i).SubItems (5).Text
148                             MetroTextBox5.Text =
149                             MetroListView1.Items(i).SubItems
150                             (6).Text
151                             MetroTextBox12.Text =
152                             MetroListView1.Items(i).SubItems
153                             (7).Text
154                             MetroListView1.Items(i).Selected = True
155                         Exit For
156                     End If
157                 Next
158             End If
159         Catch
160
161         End Try
162     End Sub
```



```
151
152         Private Sub metrobutton3_Click(ByVal sender As
System.Object, ByVal e As System.EventArgs)
153         Try
154         If MetroTextBox8.Text = "" Then
155         MsgBox("Please mention the BookID", 0, "")
156         Else
157         If objcon.State = ConnectionState.Closed Then
158         com = New OleDb.OleDbCommand("delete from
Issue where BookID=" & MetroTextBox8.Text &
159         " ", objcon)
160         If MsgBox("Do you really want to delete?",
MsgBoxStyle.YesNo, "Are you sure?") =
Windows.Forms.DialogResult.Yes Then
161         com.ExecuteNonQuery()
162         Call readData()
163         End If
164         objcon.Close()
165         End If
166         End If
167         Catch ex As Exception
168         End Try
169         End Sub
170
171         Private Sub MetroListView1_SelectedIndexChanged(ByVal
sender As
System.Object, ByVal e As System.EventArgs) Handles
MetroListView1.SelectedIndexChanged
172         Dim i As Integer
173         For i = 0 To MetroListView1.Items.Count - 1
174         If MetroListView1.Items(i).Selected = True Then
175         MetroTextBox8.Text =
MetroListView1.Items(i).SubItems(0).Text
176         Exit For
177         End If
178         Next
179         MetroListView1.Focus()
180         MetroListView1.FullRowSelect = True
```



```
181         End Sub
182
183
184
185     Private Sub MetroTextBox8_TextChanged(sender As Object, e As
186         EventArgs) Handles MetroTextBox8.TextChanged
187         Dim i As Integer
188         MetroListView1.SelectedItems.Clear()
189         MetroTextBox8.Focus()
190         Try
191             If Me.MetroTextBox8.Text = "" Then
192                 MetroTextBox6.Text = ""
193                 MetroTextBox6.Text = ""
194                 MetroTextBox2.Text = ""
195                 MetroTextBox3.Text = ""
196                 MetroTextBox4.Text = ""
197                 MetroTextBox7.Text = ""
198                 MetroTextBox5.Text = ""
199
200                 MetroTextBox12.Text = ""
201                 Else
202                 For i = 0 To MetroListView1.Items.Count - 1
203                     If MetroTextBox8.Text =
204                         MetroListView1.Items(i).SubItems
205                         (0).Text Then
206                             MetroTextBox6.Text =
207                             MetroListView1.Items(i).SubItems (1).Text
208                             MetroTextBox2.Text =
209                             MetroListView1.Items(i).SubItems (2).Text
210                             MetroTextBox3.Text =
211                             MetroListView1.Items(i).SubItems (3).Text
212                             MetroTextBox4.Text =
213                             MetroListView1.Items(i).SubItems (4).Text
214                             MetroTextBox7.Text =
215                             MetroListView1.Items(i).SubItems (5).Text
216                             MetroTextBox5.Text =
217                             MetroListView1.Items(i).SubItems
218                             (6).Text
```



```
209 MetroTextBox12.Text =  
MetroListView1.Items(i).SubItems  
(7).Text  
210 MetroListView1.Items(i).Selected = True  
211 Exit For  
212 End If  
213 Next  
214 End If  
215 Catch  
216  
217 End Try  
218 End Sub  
219  
220 Private Sub MetroTextBox9_KeyPress(sender As Object, e As  
KeyPressEventArgs) Handles MetroTextBox9.KeyPress  
221 If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not  
IsNumeric(e.KeyChar) Then  
222 MessageBox.Show("Please use RFID Scanner")  
223 e.Handled = True  
224 End If  
225 End Sub  
226  
227  
228  
229 Private Sub MetroTextBox9_TextChanged(sender As Object, e  
As EventArgs) Handles MetroTextBox9.TextChanged  
230 If MetroTextBox9.Text = "" Then  
231 MetroTextBox10.Text = ""  
232 MetroTextBox11.Text = ""  
233  
234 Else  
235  
236 Try  
237 If objcon.State = ConnectionState.Closed Then  
objcon.Open()  
238 com = New OleDb.OleDbCommand("Select B_ID,B_Name,B_Type from  
Borrowers", objcon)  
239 dr = com.ExecuteReader  
240 While dr.Read
```



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```
241             If dr.Item(0) = MetroTextBox9.Text Then
242                 MetroTextBox10.Text = dr.Item(1)
243                 MetroTextBox11.Text = dr.Item(2)
244             End If
245
246         End While
247         dr.Close()
248         objcon.Close()
249     Catch
250
251     End Try
252
253     End If
254 End Sub
255
256
257
258 Private Sub metrobutton4_Click_1(sender As Object, e As EventArgs)
259     Handles MetroButton4.Click
260     LibraryMainForm.Show()
261     Me.Close()
262     SerialPort2.Close()
263 End Sub
264
265 Private Sub metrobutton2_Click_1(sender As Object, e As EventArgs) Handles
266     MetroButton2.Click
267     If objcon.State = ConnectionState.Closed Then objcon.Open() 268
269     com = New OleDb.OleDbCommand("SELECT * FROM Issue", objcon)
270     com.Connection = objcon
271     Dim maxid As Object
272     Dim Strid As String
273     Dim intID As Integer
274
275     com.CommandText = "Select Max(Trans) as maxid from Issue"
276     maxid = com.ExecuteScalar
277
278     If maxid Is DBNull.Value Then
```



```
279         intID = 1
280         Else
281             Strid = CType(maxid, String)
282             intID = CType(Strid, String)
283             intID = intID + 1
284
285
286     End If
287
288     MetroTextBox1.Text = intID
289     objcon.Close()
290
291
292
293
294     If MetroTextBox9.Text = "" Then
295         MsgBox("Please enter the RFID Scanner!", 0, "")
296     ElseIf MetroTextBox8.Text = "" Then
297         MsgBox("Please enter the Barcode Book!", 0, "")
298     ElseIf MetroTextBox12.Text = 1 Then
299         MsgBox("You cannot borrow the last book", 0, "")
300     ElseIf MetroTextBox9.Text <> "" And MetroTextBox10.Text = ""
301         Then
302         MsgBox("Unidentified Borrower", 0, "")
303     Else
304
305     Try
306
307         '//////////////////////////////////// 308
309             If MetroTextBox8.Text = 2 Then
310                 If objcon.State = ConnectionState.Closed Then objcon.Open ()
311                 com = New OleDb.OleDbCommand("UPDATE Books SET
312                     status='Borrowed' WHERE BookID='" &
313                     MetroTextBox8.Text & "'", objcon)
314                 com.ExecuteNonQuery()
315                 objcon.Close()
316                 Call readData()
317             Else
```



```
316         If objcon.State = ConnectionState.Closed Then objcon.Open
           ()
317     com = New OleDb.OleDbCommand("UPDATE Books SET
           status='Available' WHERE BookID='" & MetroTextBox8.Text
           & "'", objcon)
318         com.ExecuteNonQuery()
319         objcon.Close()
320         Call readData()
321         End If
322
323
324     If objcon.State = ConnectionState.Closed Then objcon.Open()
325     com = New OleDb.OleDbCommand("UPDATE Books SET
Quantity='" &
           MetroTextBox12.Text - 1 & "' WHERE BookID='" &
           MetroTextBox8.Text & "'", objcon)
326         com.ExecuteNonQuery()
327         objcon.Close()
328         Call readData()
329     If objcon.State = ConnectionState.Closed Then
           objcon.Open()
330     com = New OleDb.OleDbCommand("INSERT INTO Issue
           VALUES('" & MetroTextBox8.Text & "','" &
           MetroTextBox6.Text & "','" &
           MetroTextBox2.Text & "','" & MetroTextBox9.Text & "','" &
           MetroTextBox10.Text & "','" & MetroDateTime1.Text & "','" &
           MetroDateTime2.Text & "','" & MetroTextBox1.Text & "')",
           objcon)
331         com.ExecuteNonQuery()
332         MsgBox("Book has been Issued!", 0, "")
333         Call readData()
334         objcon.Close()
335         '////////////////////
336         MetroTextBox9.Text = ""
337         MetroTextBox10.Text = ""
338         MetroTextBox11.Text = ""
339         MetroTextBox8.Text = ""
340         MetroTextBox6.Text = ""
341         MetroTextBox2.Text = ""
```



```
342         MetroTextBox3.Text = ""
343         MetroTextBox4.Text = ""
344         MetroTextBox7.Text = ""
345         MetroTextBox5.Text = ""
346         MetroTextBox12.Text = ""
347         MetroDateTime2.MinDate = DateTime.Today
348         Catch ex As Exception
349         MsgBox(ex.Message, 0, "")
350         End Try
351         End If
352
353
354     End Sub
355
356     Private Sub metrobutton1_Click(sender As Object, e As EventArgs)
357         Handles MetroButton1.Click
358         MetroTextBox9.Text = ""
359         MetroTextBox10.Text = ""
360         MetroTextBox11.Text = ""
361
362         MetroTextBox8.Text = ""
363         MetroTextBox6.Text = ""
364         MetroTextBox2.Text = ""
365         MetroTextBox3.Text = ""
366         MetroTextBox4.Text = ""
367         MetroTextBox7.Text = ""
368         MetroTextBox5.Text = ""
369         MetroTextBox12.Text = ""
370         MetroDateTime2.MinDate = DateTime.Today
371     End Sub
376 End Class
```

A.10 Library Main Form

```
1 Public Class LibraryMainForm
2
3
4     Private Sub MetroButton1_Click(sender As Object, e As EventArgs)
5         Handles MetroButton1.Click
```



```
5      AddBooks.Show()
6
7      Me.Hide()
8
9  End Sub
10
11
12
13      Private Sub MetroButton2_Click(sender As Object, e As EventArgs)
14          Handles ➤ MetroButton2.Click
15          GroupID.Show()
16          Me.Hide()
17      End Sub
18
19      Private Sub MetroButton3_Click(sender As Object, e As EventArgs)
20          Handles ➤ MetroButton3.Click
21          BookDetail.Show()
22          Me.Hide()
23      End Sub
24
25  Private Sub Button4_Click(sender As Object, e As EventArgs) 24
26  End Sub
27
28      Private Sub MetroButton4_Click(sender As Object, e As EventArgs)
29          Handles ➤ MetroButton4.Click
30          AddBorrower.Show()
31          Me.Hide()
32      End Sub
33
34      Private Sub MetroButton5_Click(sender As Object, e As EventArgs)
35          Handles ➤ MetroButton5.Click
36          BorrowerDetail.Show()
37          Me.Hide()
38      End Sub
39
40      Private Sub MetroButton6_Click(sender As Object, e As EventArgs)
41          Handles ➤ MetroButton6.Click
42          ReturnBook.Show()
43          Me.Hide()
44      End Sub
```



```
40         End Sub
41
42     Private Sub MetroButton7_Click(sender As Object, e As EventArgs) Handles MetroButton7.Click
43         IssueBook.Show()
44         Me.Hide()
45     End Sub
46
47     Private Sub MainForm_Load(sender As Object, e As EventArgs) Handles MyBase.Load
48
49     End Sub
50
51     Private Sub MetroButton9_Click(sender As Object, e As EventArgs) Handles MetroButton9.Click
52         Try
53             If MessageBox.Show("Are you sure you want to Log out?", "",
54                               MessageBoxButtons.YesNo) = Windows.Forms.DialogResult.Yes
55             Then
56                 If Windows.Forms.DialogResult.Yes Then
57                     MessageBox.Show("Thank You!", "",
58                                   MessageBoxButtons.OK)
59                     MainForm.Show()
60                     Me.Close()
61                 End If
62             End If
63         Catch
64
65         End Try
66     End Sub
67
68     Private Sub MetroButton10_Click(sender As Object, e As EventArgs)
69
70
71 End Sub
```



```
72
73     Private Sub MetroButton8_Click(sender As Object, e As EventArgs)
74         AllBorrowed.Show()
75         Me.Hide()
76
77
78 End Sub
79
80     Private Sub MetroButton10_Click_1(sender As Object, e As EventArgs)
81         Handles MetroButton10.Click
82         Archive.Show()
83         Me.Hide()
84 End Sub
85
86     Private Sub MetroButton8_Click_1(sender As Object, e As EventArgs)
87         Handles MetroButton8.Click
88         LogBook.Show()
89         Me.Hide()
90 End Sub
91 End Class
92
```

A.11 Log Book

```
2
3     Private Sub MetroButton4_Click(sender As Object, e As EventArgs) Handles
1 Public Class LogBook
    MetroButton4.Click
4         LibraryMainForm.Show()
5         Me.Close()
6     End Sub
7
8     Sub readData2()
9         ListView2.Clear()
10        ListView2.Columns.Add("Name", 90, HorizontalAlignment.Center)
11        ListView2.Columns.Add("Address", 170, HorizontalAlignment.Center)
12        ListView2.Columns.Add("Type", 90, HorizontalAlignment.Center)
```



```
13 ListView2.Columns.Add("Time", 70, HorizontalAlignment.Center)
14 ListView2.Columns.Add("Action", 90, HorizontalAlignment.Center)
15 ListView2.Columns.Add("Date", 80, HorizontalAlignment.Center)
16 ListView2.View = View.Details
17 Try
18
19     If (objcon.State = ConnectionState.Closed) Then objcon.Open()
20     com = New OleDb.OleDbCommand("SELECT A_Name,
21     A_Address, A_Type, A_Time, A_Action, A_Date FROM
22     Attendance WHERE A_Date = DATE()", objcon)
23     dr = com.ExecuteReader
24     While dr.Read()
25     Call adddatatolistview2(ListView2, dr(0), dr(1), dr(2), dr(3),
26     dr(4), dr(5))
27     End While
28     dr.Close()
29     objcon.Close()
30     Catch
31     'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
32     End Try
33     End Sub
```

```
34 Public Sub adddatatolistview2(ByVal lvw
35 As ListView,
36 ByVal A_Name As String,
37 ByVal A_Address As String,
38 ByVal A_Type As String,
39 ByVal A_Time As String,
40 ByVal A_Action As String,
41 ByVal A_Date As String)
42 Dim lv As New ListViewItem
43 lvw.Items.Add(lv)
44 lv.Text = A_Name
45 lv.SubItems.Add(A_Address)
46 lv.SubItems.Add(A_Type)
47 lv.SubItems.Add(A_Time)
48 lv.SubItems.Add(A_Action)
49 lv.SubItems.Add(A_Date)
50 End Sub
```



```
48
49     Private Sub LogBook_Load(sender As Object, e As EventArgs) Handles
        MyBase.Load
50     Call readData2()
51     MetroDateTime1.MaxDate = DateTime.Today
52     End Sub
53
54     Private Sub MetroDateTime1_ValueChanged(sender As Object, e As EventArgs)
        Handles MetroDateTime1.ValueChanged
55     ListView2.Clear()
56     ListView2.Columns.Add("Name", 90, HorizontalAlignment.Center)
57     ListView2.Columns.Add("Address", 170, HorizontalAlignment.Center)
58     ListView2.Columns.Add("Type", 90, HorizontalAlignment.Center)
59     ListView2.Columns.Add("Time", 70, HorizontalAlignment.Center)
60     ListView2.Columns.Add("Action", 90, HorizontalAlignment.Center)
61     ListView2.Columns.Add("Date", 80, HorizontalAlignment.Center)
62     ListView2.View = View.Details
63     Try
64
65         If (objcon.State = ConnectionState.Closed) Then objcon.Open()
66         com = New OleDb.OleDbCommand("SELECT A_Name,
            A_Address, A_Type,
            A_Time, A_Action, A_Date FROM Attendance WHERE A_Date = '" &
            MetroDateTime1.Text & "'", objcon)
67
68         dr = com.ExecuteReader
69         While dr.Read()
70             Call adddatatolistview2(ListView2, dr(0), dr(1), dr(2), dr(3),
71             dr(4), dr(5))
72         End While
73         dr.Close()
74         objcon.Close()
75         Catch
76             'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
77         End Try
78     End Sub
79 End Class
```

A.13 Login Form



```
1 Public Class LoginForm
2
3
4 Private Sub BtnClearM_Click(sender As Object, e As EventArgs) Handles
    BtnClearM.Click
5
6     UserNameTextBox2M.Text = ""
7     PassWordTextBox2M.Text = ""
8     UserNameTextBox2M.Focus()
9     MainForm.Show()
10    Me.Close()
11 End Sub
12
13 Private Sub BtnLogin2M_Click(sender As Object, e As EventArgs)
    Handles BtnLogin2M.Click
14 Dim PassWord2 As String
15
16 Dim UserName2 As String
17
18 Try
19
20
21     If (objcon.State = ConnectionState.Closed) Then objcon.Open()
22     com = New OleDb.OleDbCommand("SELECT UserName, PassWord
        FROM
        Login WHERE (UserName = " & UserNameTextBox2M.Text & " ) AND
        (PassWord = " & PassWordTextBox2M.Text & ")", objcon)
23     dr = com.ExecuteReader
24     If dr.HasRows Then 25 While dr.Read()
26
27         'Do something here
28         PassWord2 = dr("PassWord").ToString()
29         UserName2 = dr("UserName").ToString()
30
31
32     If PassWord2 = PassWordTextBox2M.Text And UserName2 =
        UserNameTextBox2M.Text Then
33
34         MessageBox.Show("Logged in successfully as " &
```

➤

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```
        UserName2, "", MessageBoxButtons.OK,
        MessageBoxIcon.Information)
35        LibraryMainForm.Show()
36        Me.Hide()
37        'Clear all fields
38        PassWordTextBox2M.Text = ""
39        UserNameTextBox2M.Text = ""
40
41        Else
42
43        End If
44
45    End While
46        ElseIf PassWordTextBox2M.Text = "" And
        UserNameTextBox2M.Text = "" Then
47            MessageBox.Show("Please input details", "Authentication
        Failure", MessageBoxButtons.OK, MessageBoxIcon.Exclamation)
48
49
50        Else
51            MessageBox.Show("Incorrect Username and Password",
        "Authentication Failure", MessageBoxButtons.OK,
        MessageBoxIcon.Exclamation)
52            PassWordTextBox2M.Text = ""
53            UserNameTextBox2M.Text = ""
54        End If
55
56
57    Catch ex As Exception
58        'MessageBox.Show("Error while connecting to SQL Server." & ex.Message)
59    Finally
60        objcon.Close() 'Whether there is error or not. Close the connection.
61
62    End Try 63 End Sub
64
65
66
67 Private Sub LoginForm_Load(sender As Object, e As EventArgs) Handles
        MyBase.Load
```



```
68
69 End Sub
70
71
72
73
74
75
76
77
78
79 Private Sub UserNameTextBox2M_KeyPress1(sender As Object, e As
    KeyPressEventArgs) Handles UserNameTextBox2M.KeyPress
80
104
105 End Sub
106
107 Private Sub UserNameTextBox2M_Click(sender As Object, e As EventArgs)
    Handles UserNameTextBox2M.Click
108
109 End Sub 110 End Class
111
112
```

A.14 MainForm

```
2
3 Private Sub MetroButton1_Click(sender As Object, e As EventArgs) Handles
1 Public Class MainForm
    MetroButton1.Cli
ck 4
LoginForm.Show()
5 Me.Hide()
6 LoginForm.UserNameTextBox2M.Focus()
7
8 End Sub
9
```



```
10 Private Sub MetroButton2_Click(sender As Object, e As EventArgs)
    Handles ➤ MetroButton2.Click
    AttendanceMainForm.Show()
11
12
13
14 Me.Hide()
15 End Sub
16
17 Private Sub MetroButton3_Click(sender As Object, e As EventArgs)
    Handles ➤ MetroButton3.Click
    SearchMainForm.Show()
18
19 Me.Hide()
20 End Sub
21
22 Private Sub MetroButton4_Click(sender As Object, e As EventArgs)
23 Me.Hide()
24 Me.Close()
25 End Sub
26
27 Private Sub MainForm_Load(sender As Object, e As EventArgs) Handles ➤
    MyBase.Load
28
29 End Sub
30 End Class
```

A.15 Return Book

```
1 Imports System
2 Imports System.IO.Ports 3
4 Public Class ReturnBook
5
6
7
8
9
10 Sub ClearThem()
11
12
13 MetroTextBox8.Text = ""
14 MetroTextBox6.Text = ""
```



```
15 MetroTextBox2.Text = ""
16 MetroTextBox3.Text = ""
17 MetroTextBox4.Text = ""
18 MetroTextBox7.Text = ""
19 MetroTextBox5.Text = ""
20 'metrotextbox9.text = ""
21 MetroTextBox1.Text = ""
22 ComboBox1.Items.Clear()
23 MetroTextBox10.Text = ""
24 MetroTextBox11.Text = ""
25 MetroDateTime2.Text = ""
26 MetroTextBox13.Text = ""
27 MetroTextBox14.Text = ""
28
29 MetroDateTime1.MinDate = DateTime.Today
30 End Sub
31
32
33
34 Sub BookID_Combo()
35 Try
36 If objcon.State = ConnectionState.Closed Then objcon.Open()
37 com = New OleDb.OleDbCommand("Select BookID from
38 Books", objcon)
39 dr = com.ExecuteReader
40 While dr.Read
41 ComboBox1.Items.Add(dr.Item(0))
42 End While
43 dr.Close()
44 objcon.Close()
45 Catch ex As Exception
46
47 End Try
48 End Sub
49
50 Private Sub ReturnBook_Load(ByVal sender As System.Object, ByVal e
51 As System.EventArgs) Handles MyBase.Load
52 Call BookID_Combo()
```



```
52      Call readData()  
53      Call readData2()  
54      ' MetroDateTime1.MinDate = DateTime.Today  
55      MetroDateTime1.Enabled = False  
56  
57  
58  
59 End Sub  
60  
61      Sub readData2()  
62      ListView2.Clear()  
63      ListView2.Columns.Add("Book Barcode", 160,  
64      HorizontalAlignment.Center)  
65      ListView2.Columns.Add("Group ID", 60, HorizontalAlignment.Center)  
66      ListView2.Columns.Add("Book Name", 310, HorizontalAlignment.Center)  
67      ListView2.Columns.Add("Borrower ID", 90, HorizontalAlignment.Center)  
68      ListView2.Columns.Add("Borrower Name", 190,  
69      HorizontalAlignment.Center)  
70      ListView2.Columns.Add("Issue Date", 90, HorizontalAlignment.Center)  
71      ListView2.Columns.Add("Due Date", 90, HorizontalAlignment.Center)  
72      ListView2.Columns.Add("Transaction", 130, HorizontalAlignment.Center)  
73      ListView2.View = View.Details  
74      Try  
75          If (objcon.State = ConnectionState.Closed) Then objcon.Open()  
76          com = New OleDb.OleDbCommand("SELECT * FROM Issue",  
77          objcon)  
78          dr = com.ExecuteReader  
79          While dr.Read()  
80              Call adddatatolistview2(ListView2, dr(0), dr(1), dr(2), dr(3),  
81              dr(4), dr(5), dr(6), dr(7))  
82          End While  
83          dr.Close()  
84          objcon.Close()  
85          Catch  
86              MsgBox("Please Refresh", MsgBoxStyle.Information, "")  
87          End Try  
88      End Sub
```



```
87      Public Sub adddatatolistview2(ByVal lvw As
88          ListView,
89          ByVal BookID As String,
90          ByVal GroupID As String,
91          ByVal BookName As String,
92          ByVal B_ID As String,
93          ByVal B_Name As String,
94          ByVal Issue_Date As String,
95          ByVal Due_Date As String,
96          ByVal Trans As String)
97      Dim lv As New ListViewItem
98      lvw.Items.Add(lv)
99      lv.Text = BookID
100     lv.SubItems.Add(GroupID)
101     lv.SubItems.Add(BookName)
102     lv.SubItems.Add(B_ID)
103     lv.SubItems.Add(B_Name)
104     lv.SubItems.Add(Issue_Date)
105     lv.SubItems.Add(Due_Date)
106     lv.SubItems.Add(Trans)
107     End Sub
108
109
110     Private Sub MetroButton1_Click(ByVal sender As System.Object, ByVal e
111         As System.EventArgs) Handles MetroButton1.Click
112         Me.MetroTextBox8.Text = ""
113         MetroTextBox6.Text = ""
114         MetroTextBox2.Text = ""
115         MetroTextBox3.Text = ""
116         MetroTextBox4.Text = ""
117         MetroTextBox7.Text = ""
118         MetroTextBox5.Text = ""
119         MetroTextBox1.Text = ""
120         'metrottextbox9.text = ""
121         MetroTextBox10.Text = ""
122         MetroTextBox11.Text = ""
123         MetroDateTime2.Text = ""
```



```
124     MetroTextBox13.Text = ""
125     End Sub
126
127
128
129
130
131
132 Private Sub ListView2_SelectedIndexChanged(sender As Object, e As EventArgs)
    Handles ListView2.SelectedIndexChanged
133
134
135
136         Dim i As Integer
137         For i = 0 To ListView2.Items.Count - 1
138             If ListView2.Items(i).Selected = True Then
139                 MetroTextBox8.Text =
                    ListView2.Items(i).SubItems(0).Text
140                 ComboBox1.Text = ListView2.Items(i).SubItems(0).Text
141                 MetroTextBox11.Text = ListView2.Items(i).SubItems(3).Text
142                 MetroTextBox10.Text = ListView2.Items(i).SubItems(4).Text
143                 'metrotextbox9.text = ListView2.Items(i).SubItems(5).Text
144                 MetroLabel9.Text = ListView2.Items(i).SubItems(6).Text
145                 MetroTextBox14.Text = ListView2.Items(i).SubItems(7).Text
146
147
148                 Exit For
149             End If
150         Next
151         ListView2.Focus()
152         ListView2.FullRowSelect = True
153         MetroTextBox12.Text = 0
154         MetroTextBox15.Text = 0
155     End Sub
156
157
158 Private Sub MetroButton2_Click(sender As Object, e As EventArgs)
    Handles MetroButton2.Click
159     Dim dt1 As DateTime =
        Convert.ToDateTime(MetroDateTime2.Text)
```



```
160      Dim dt2 As DateTime = Convert.ToDateTime(MetroDateTime1.Text)
161      Dim ts As TimeSpan = dt2.Subtract(dt1) 162
163      Dim Total As Double = 0.0
164      Dim Penalty As Integer = 15 165      Dim Damage
      As Integer = 150
166      If dt1 < dt2 Then
167          Total = Penalty * ts.Days
168
169      MetroTextBox12.Text = Total
170
171  End If
172
173      If ComboBox1.Text = "" Then
174          MsgBox("Please mention the Book ID", 0, "")
175      Else
176          Try
177
178
179      If MessageBox.Show("Damage book or not?", "",
      MessageBoxButtons.YesNo) =
      Windows.Forms.DialogResult.Yes Then
180
181
182          MetroTextBox15.Text = 150
183          If objcon.State = ConnectionState.Closed Then objcon.Open
      ()
184          com = New OleDb.OleDbCommand("INSERT INTO
      DamageBooks VALUES(" & MetroTextBox8.Text &
      ", " &
      MetroTextBox2.Text & ", " & MetroTextBox11.Text
      & ", " & MetroTextBox10.Text & ", " & MetroLabel9.Text
      & ", " & MetroDateTime2.Text & ", " &
      MetroDateTime1.Text & ", " &
      MetroTextBox15.Text & ")", objcon)
185          com.ExecuteNonQuery()
186          objcon.Close()
187          Call readData()
188
189
```



```
190
191
192     If objcon.State = ConnectionState.Closed Then objcon.Open
        ()
193     com = New OleDb.OleDbCommand("DELETE FROM Issue
        WHERE Trans=" & MetroTextBox14.Text & "", objcon)
194     com.ExecuteNonQuery()
195     objcon.Close()
196     Call readData()
197     Call readData2()
198
199
200     If MetroTextBox8.Text = 2 Then
201     If objcon.State = ConnectionState.Closed Then
        objcon.Open()
202     com = New OleDb.OleDbCommand("UPDATE Books
        SET status='Available' WHERE BookID=" &
        MetroTextBox8.Text & "", objcon)
203     com.ExecuteNonQuery()
204     objcon.Close()
205     Call readData()
206     Call readData2()
207     Else
208     If objcon.State = ConnectionState.Closed Then
        objcon.Open()
209     com = New OleDb.OleDbCommand("UPDATE Books
        SET status='Borrowed' WHERE BookID=" &
        MetroTextBox8.Text & "", objcon)
210     com.ExecuteNonQuery()
211     objcon.Close()
212     Call readData()
213     Call readData2()
214     End If
215
216
217     If objcon.State = ConnectionState.Closed Then objcon.Open
        ()
218     com = New OleDb.OleDbCommand("INSERT INTO Returns
        VALUES
```



```
219 com.ExecuteNonQuery()  
220 MsgBox("Book has been returned!", 0, "")  
221 objcon.Close()  
222 Call readData2()  
223  
224 Call ClearThem()  
225 MsgBox(Damage, 0, "Damage Fine")  
226 MsgBox(Total, 0, "Late Penalty")  
227 Else  
228  
229 If objcon.State = ConnectionState.Closed Then objcon.Open  
230 ()  
231 com = New OleDb.OleDbCommand("DELETE FROM Issue  
232 WHERE Trans=" & MetroTextBox14.Text & "", objcon)  
233 com.ExecuteNonQuery()  
234 objcon.Close()  
235 Call readData()  
236 Call readData2()  
237  
238 If MetroTextBox8.Text > 2 Then  
239 If objcon.State = ConnectionState.Closed Then  
240 objcon.Open()  
241 com = New OleDb.OleDbCommand("UPDATE Books  
242 SET status='Available' WHERE BookID=" &  
243 MetroTextBox8.Text & "", objcon)  
244 com.ExecuteNonQuery()  
245 objcon.Close()  
Call readData()  
Call readData2()  
Else  
If objcon.State = ConnectionState.Closed Then  
objcon.Open()  
com = New OleDb.OleDbCommand("UPDATE Books  
SET
```



```
status='Borrowed' WHERE BookID="" & MetroTextBox8.Text &
    "", objcon)
246         com.ExecuteNonQuery()
247         objcon.Close()
248         Call readData()
249         Call readData2()
250         End If
251
252
253     If objcon.State = ConnectionState.Closed Then objcon.Open
        ()
254     com = New OleDb.OleDbCommand("UPDATE Books SET
        Quantity="
        & MetroTextBox13.Text + 1 & " WHERE BookID="
        & MetroTextBox8.Text & "", objcon)
255     com.ExecuteNonQuery()
256     objcon.Close()
257     Call readData()
258     Call readData2()
259
260     If objcon.State = ConnectionState.Closed Then objcon.Open
        ()
261     com = New OleDb.OleDbCommand("INSERT INTO Returns
        VALUES
        (" & MetroTextBox8.Text & "," & MetroTextBox2.Text &
        "," & MetroTextBox11.Text & "," & MetroTextBox10.Text
        & "," & MetroLabel9.Text & "," & MetroDateTime2.Text &
        "," & MetroDateTime1.Text & "," & MetroTextBox12.Text
        & ")", objcon)
262     com.ExecuteNonQuery()
263     MsgBox("Book has been returned!", 0, "")
264     objcon.Close()
265     Call readData2()
266
267     Call ClearThem()
268
269
270     MsgBox(Total, 0, "Late Penalty")
271     End If
```



```
272
273
274         MetroTextBox15.Text = ""
275         MetroTextBox12.Text = ""
276
277
278         Catch ex As Exception
279             MsgBox(ex.Message, 0, "")
280         End Try
281
282
283     End If
284 End Sub
285
286 Private Sub MetroButton4_Click(sender As Object, e As EventArgs)
287     Handles MetroButton4.Click
288         LibraryMainForm.Show()
289         Me.Close()
290 End Sub
291
292
293
294
295 Sub IssueDetail() '
296 Try
297 If objcon.State = ConnectionState.Closed Then objcon.Open()
298 com = New OleDb.OleDbCommand("Select IssueDate,
Barcode_ID,
Borrower_Name, DueDate from Issue WHERE BookID='" &
MetroTextBox8.Text & "'", objcon)
299 dr = com.ExecuteReader
300 While dr.Read
301 MetroLabel9.Text = dr.Item(0)
302 MetroTextBox11.Text = dr.Item(1)
303 MetroTextBox10.Text = dr.Item(2)
304 MetroDateTime2.Text = dr.Item(3)
305 MetroTextBox14.Text = dr.Item(7)
306 End While
```



```
307         dr.Close()
308         objcon.Close()
309         Catch ex As Exception
310
311     End Try
312 End Sub
353 Private Sub MetroTextBox8_TextChanged(sender As Object, e As EventArgs)
    Handles MetroTextBox8.TextChanged
354
355
356     Dim i As Integer
357     MetroListView2.SelectedItems.Clear()
358     MetroTextBox8.Focus()
359     Try
360     If Me.MetroTextBox8.Text = "" Then
361         MetroTextBox6.Text = ""
362         MetroTextBox6.Text = ""
363         MetroTextBox2.Text = ""
364         MetroTextBox3.Text = ""
365         MetroTextBox4.Text = ""
366         MetroTextBox7.Text = ""
367         MetroTextBox5.Text = ""
368         MetroTextBox1.Text = ""
369
370         MetroTextBox13.Text = ""
371     Else
372     For i = 0 To MetroListView2.Items.Count - 1
373         If MetroTextBox8.Text =
            MetroListView2.Items(i).SubItems
374             (0).Text Then
            ComboBox1.Text =
            MetroListView2.Items(i).SubItems
375             (0).Text
            MetroTextBox6.Text =
            MetroListView2.Items(i).SubItems (1).Text
376             MetroTextBox2.Text =
            MetroListView2.Items(i).SubItems (2).Text
377             MetroTextBox3.Text =
            MetroListView2.Items(i).SubItems (3).Text
```



```
378 MetroTextBox4.Text =  
MetroListView2.Items(i).SubItems (4).Text  
379 MetroTextBox7.Text =  
MetroListView2.Items(i).SubItems (5).Text  
380 MetroTextBox5.Text =  
MetroListView2.Items(i).SubItems  
(6).Text  
381 MetroTextBox13.Text =  
MetroListView2.Items(i).SubItems  
(7).Text  
382 MetroListView2.Items(i).Selected = True  
383 Exit For  
384 End If  
385 Next  
386 End If  
387 Catch  
388  
389 End Try  
390 Call IssueDetail()  
391 End Sub  
392  
393  
416  
417 Sub readData()  
418 MetroListView2.Clear()  
419 MetroListView2.Columns.Add("Book Barcode", 160,  
HorizontalAlignment.Center)  
420 MetroListView2.Columns.Add("Group ID", 60,  
HorizontalAlignment.Center)  
421 MetroListView2.Columns.Add("Book Name", 310,  
HorizontalAlignment.Center)  
422 MetroListView2.Columns.Add("Publisher", 90,  
HorizontalAlignment.Center)  
423 MetroListView2.Columns.Add("Author", 90, HorizontalAlignment.Center)  
424 MetroListView2.Columns.Add("Publishing Year", 190,  
HorizontalAlignment.Center)  
425 MetroListView2.Columns.Add("Edition", 90, HorizontalAlignment.Center)  
426 MetroListView2.Columns.Add("Quantity", 90,  
HorizontalAlignment.Center)
```



```
427 MetroListView2.View = View.Details
428 Try
429
430     If (objcon.State = ConnectionState.Closed) Then
431         objcon.Open()
432         com = New OleDb.OleDbCommand("SELECT * FROM
433         Books", objcon)
434         dr = com.ExecuteReader
435         While dr.Read()
436             Call adddatatolistview(MetroListView2, dr(0), dr(1), dr(2), dr
437             (3), dr(4), dr(5), dr(6), dr(7))
438         End While
439     End Try
440     End Sub
441
442
443     Public Sub adddatatolistview(ByVal lvw As ListView, ByVal BookID As
444     String, ByVal GroupID As String, ByVal BookName As String, ByVal
445     Publisher As String, ByVal Author As String, ByVal PubYear As String,
446     ByVal edi As String, ByVal qty As String)
447         Dim lv As New ListViewItem
448         lvw.Items.Add(lvw)
449         lv.Text = BookID
450         lv.SubItems.Add(GroupID)
451         lv.SubItems.Add(BookName)
452         lv.SubItems.Add(Publisher)
453         lv.SubItems.Add(Author)
454         lv.SubItems.Add(PubYear)
455         lv.SubItems.Add(edi)
456         lv.SubItems.Add(qty)
457     End Sub
458 End Class
```

A.17 Search Main Form

```
1 Public Class SearchMainForm
2
```



```
3 Dim sel As Integer
4 Private Sub metrocombobox1_SelectedIndexChanged(ByVal sender As
    System.Object, ByVal e As System.EventArgs) Handles
    MetroComboBox1.SelectedIndexChanged
5
6
7 If MetroComboBox1.Text = "Status" Then
8 MetroComboBox2.Enabled = True
9 MetroComboBox2.Visible = True 10 MetroTextBox1.Visible = False
11
12 Else
13 MetroComboBox2.Enabled = False
14 MetroComboBox2.Visible = False
15 MetroTextBox1.Visible = True
16
17 End If
18 Call forselect()
19 End Sub
20 Sub forselect()
21 If MetroComboBox1.Text = "Barcode" Then
22 sel = 1
23 ElseIf MetroComboBox1.Text = "Name" Then
24 sel = 2
25 ElseIf MetroComboBox1.Text = "Author" Then
26 sel = 3
27 ElseIf MetroComboBox1.Text = "Status" Then
28 sel = 8
29 ElseIf MetroComboBox1.Text = "Category" Then
30 sel = 9
31 'ElseIf MetroComboBox1.Text = "Publishing-Year" Then
32 ' sel = 6
33
34 End If
35 End Sub
36
37 Private Sub BookDetail_Load(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles MyBase.Load
38 metrocombobox2.Visible = False
39 metrotextbox1.Visible = False
```



```
40
41     Call readData()
42     Call readData2()
43     End Sub
44
45
46     Sub readData2()
47         MetroListView2.Clear()
48         MetroListView2.Columns.Add("Book Barcode", 90)
49         MetroListView2.Columns.Add("Category", 90)
50         MetroListView2.Columns.Add("Book Name", 90)
51         MetroListView2.Columns.Add("Publisher", 90)
52         MetroListView2.Columns.Add("Author", 90)
53         MetroListView2.Columns.Add("Publishing Year", 90)
54         MetroListView2.Columns.Add("Edition", 90)
55         MetroListView2.Columns.Add("Available Copies", 90)
56         MetroListView2.View = View.Details
57     Try
58
59         If (objcon.State = ConnectionState.Closed) Then objcon.Open()
60         com = New OleDb.OleDbCommand("SELECT * FROM Books",
61         objcon)
62         dr = com.ExecuteReader
63         While dr.Read()
64             Call adddatatolistview(MetroListView2, dr(0), dr(1), dr(2), dr
65             (3), dr(4), dr(5), dr(6), dr(7))
66         End While
67         dr.Close()
68         objcon.Close()
69     Catch
70         MsgBox("Please Refresh", MsgBoxStyle.Information, "")
71     End Try
72     End Sub
73
74
75     Sub readData()
76         MetroListView2.Clear()
```



```
77 MetroListView2.Columns.Add("Book Barcode", 90)
78 MetroListView2.Columns.Add("Category", 90)
79 MetroListView2.Columns.Add("Book Name", 90)
80 MetroListView2.Columns.Add("Publisher", 90)
81 MetroListView2.Columns.Add("Author", 90)
82 MetroListView2.Columns.Add("Publishing Year", 90)
83 MetroListView2.Columns.Add("Edition", 90)
84 MetroListView2.Columns.Add("Available Copies", 90)
85
86
87 MetroListView2.View = View.Details
88 sel = 5
89 'Call whenclick()
90 End Sub
91 Sub whenclick()
92 Try
93
94     While dr.Read()
95         Call adddatatolistview(MetroListView2, dr(0), dr(1), dr(2), dr
            (3), dr(4), dr(5), dr(6), dr(7))
96     End While
97     dr.Close()
98     objcon.Close()
99     Catch
100     'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
101     End Try
102     End Sub
103     Public Sub adddatatolistview(ByVal lvw As ListView, ByVal
        BookID As String, ByVal GroupID As String, ByVal BookName
        As String, ByVal publisher As
String, ByVal author As String, ByVal pubyear As String, ByVal edi As
String, ByVal qty As String)
104         Dim lv As New ListViewItem
105         lvw.Items.Add(lv)
106         lv.Text = BookID
107         lv.SubItems.Add(GroupID)
108         lv.SubItems.Add(BookName)
109         lv.SubItems.Add(publisher)
110         lv.SubItems.Add(author)
```



```
111         lv.SubItems.Add(pubyear)
112         lv.SubItems.Add(edi)
113         lv.SubItems.Add(qty)
114     End Sub
115
116 Private Sub metrobutton1_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles MetroButton1.Click
117
118
119     Dim SearchFilter = MetroTextBox1.Text
120     Dim SearchFilter2 = MetroComboBox2.Text
121
122
123     If objcon.State = ConnectionState.Closed Then objcon.Open() 124
125         Select Case (sel)
126         Case 1
127             com = New OleDb.OleDbCommand("select * from Books where
                BookID LIKE '%" & SearchFilter & "%' AND Quantity > 1", objcon)
128                 dr = com.ExecuteReader
129
130
131             If dr.HasRows Then
132                 Call readData()
133                 Call whenclick()
134                 Call forselect()
135                 MetroTabControl2.Visible = True
136             Else
137                 MsgBox("No Data")
138                 MetroTabControl2.Visible = False
139             End If
140
141
142
143         Case 2
144             com = New OleDb.OleDbCommand("select * from Books where
                BookName
145                 LIKE '%" & SearchFilter & "%' AND Quantity > 1", objcon)
146                 dr = com.ExecuteReader
```



```
147
148
149         If dr.HasRows Then
150             Call readData()
151             Call whenclick()
152             Call forselect()
153             MetroTabControl2.Visible = True
154         Else
155             MsgBox("No Data")
156             MetroTabControl2.Visible = False
157         End If
158
159
160
161         Case 3
162         com = New OleDb.OleDbCommand("select * from Books
163             where Author LIKE '%" & SearchFilter & "%' AND Quantity
164             > 1", objcon)
165         dr = com.ExecuteReader
166         If dr.HasRows Then
167             Call readData()
168             Call whenclick()
169             Call forselect()
170             MetroTabControl2.Visible = True
171         Else
172             MsgBox("No Data")
173             MetroTabControl2.Visible = False
174         End If
175
176         Case 8
177
178         If MetroComboBox2.Text = "Available" Then
179             com = New OleDb.OleDbCommand("select * from
180             Books where Quantity>1", objcon)
181             dr = com.ExecuteReader
182             If dr.HasRows Then
183                 Call readData()
184                 Call whenclick()
```



```
183         Call forselect()
184         MetroTabControl2.Visible = True
185     Else
186         MsgBox("No Data")
187         MetroTabControl2.Visible = False
188     End If
189
190
191     ElseIf MetroComboBox2.Text = "Not Available" Then
192         com = New OleDb.OleDbCommand("select * from
193         Books where Quantity =1", objcon)
194         dr = com.ExecuteReader
195         If dr.HasRows Then
196             Call readData()
197             Call whenclick()
198             Call forselect()
199             MetroTabControl2.Visible = True
200         Else
201             MsgBox("No Data")
202             MetroTabControl2.Visible = False
203         End If
204
205     End If
206
207
208     Case 9
209
210         com = New OleDb.OleDbCommand("select * from Books
211         where GroupID LIKE '%" & SearchFilter & "%' AND
212         Quantity > 1", objcon)
213         dr = com.ExecuteReader
214         If dr.HasRows Then
215             Call readData()
216             Call whenclick()
217             Call forselect()
218             MetroTabControl2.Visible = True
219         Else
220             MsgBox("No Data")
```



```
219 MetroTabControl2.Visible = False
220 End If
221
222
223 Case 6
224
225 com = New OleDb.OleDbCommand("select * from Books
where pubyear LIKE '%" & MetroTextBox1.Text & "%' AND
Quantity > 1", objcon)
226 dr = com.ExecuteReader
227 If dr.HasRows Then
228 Call readData()
229 Call whenclick()
230 Call forselect()
231 MetroTabControl2.Visible = True
232 Else
233 MsgBox("No Data")
234 MetroTabControl2.Visible = False
235 End If
236
237
238 End Select
239
240 objcon.Close()
250 End Sub
251
252 Private Sub MetroListView2_SelectedIndexChanged(ByVal sender As
System.Object, ByVal e As System.EventArgs)
253
254 End Sub
255
256 Private Sub Button6_Click(ByVal sender As System.Object,
ByVal e As System.EventArgs)
257 Try
258 Dim i As Integer
259 For i = 0 To MetroListView2.Items.Count - 1
260 If MetroListView2.Items(i).Selected = True Then
261 metrottextbox1.Text = MetroListView2.Items(i + 1).SubItems
(0).Text
```



```
262         Exit For
263     End If
264 Next
265 MetroListView2.Focus()
266 MetroListView2.FullRowSelect = True
267 Catch ex As Exception
268
269 End Try
270 End Sub
271
272 Private Sub Button5_Click(ByVal sender As System.Object,
273     ByVal e As System.EventArgs)
274     Try
275         Dim i As Integer
276         For i = 0 To MetroListView2.Items.Count - 1
277             If MetroListView2.Items(i).Selected = True Then
278                 metrotextbox1.Text = MetroListView2.Items(i - 1).SubItems
279                 (0).Text
280             Exit For
281         End If
282     Next
283     MetroListView2.Focus()
284     MetroListView2.FullRowSelect = True
285     Catch ex As Exception
286
287 End Try
288 End Sub
289
290 Private Sub Button1_Click(sender As Object, e As EventArgs)
291     LibraryMainForm.Show()
292     Me.Hide()
293 End Sub
294
295 Private Sub MetroButton2_Click(sender As Object, e As EventArgs)
296     Handles MetroButton2.Click
297     MainForm.Show()
298     Me.Close()
```



298 End Sub

299

300

301 End Class

A.18 Arduino Code

//This code was adapted from the ChangeUID example code from the MFRC522 libraries.

```
#include <SPI.h>
```

```
#include <MFRC522.h>
```

```
#define RST_PIN 9
```

```
#define SS_PIN 10
```

```
const int buzzer = 4; //buzzer to arduino pin 4
```

```
MFRC522 mfrc522(SS_PIN, RST_PIN); // Create MFRC522 instance MFRC522::MIFARE_Key key;
```

```
void setup() {
```

```
    // put your setup code here, to run once:
```

```
    Serial.begin(9600); // Initialize serial communications with the PC while (!Serial); // Do  
    nothing if no serial port is opened (added for
```

```
    Arduinos based on ATMEGA32U4)
```

```
    SPI.begin(); // Init SPI bus mfrc522.PCD_Init(); // Init  
    MFRC522 card
```

```
    Serial.println(F("Warning: this example overwrites the UID of your UID changeable card, use  
    with care!"));
```

```
}
```

```
void loop() {
```



// put your main code here, to run repeatedly:

```
// Look for new cards, and select one if present  if ( ! mfrc522.PICC_IsNewCardPresent() || !  
mfrc522.PICC_ReadCardSerial() ) {  delay(1500);  return; }
```

// Now a card is selected. The UID and SAK is in mfrc522.uid.

```
//Serial.print(F("Card UID:"));
```

```
for (byte i = 0; i < mfrc522.uid.size; i++) {
```

```
  Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
```

```
  Serial.print(mfrc522.uid.uidByte[i], HEX);
```

```
}  tone(buzzer, 4000); // Send 1KHz sound signal...
```

```
delay(100);    // ...for 1 sec
```

```
noTone(buzzer); // Stop
```

```
sound... delay(100);    //
```

```
...for 1sec
```

```
//Serial.println();
```

```
}
```



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B. RESEARCHER'S WEEKLY PROGRESS REPORTS

RESEARCHER'S WEEKLY PROGRESS REPORTS

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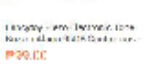
ECE 201:DESIGN PROJECT 2 WEEKLY PROGRESS REPORT

PROJECT TITLE:	Automated Library Management System using Barcode and RFID
-----------------------	---

PROPONENTS:	Melvin Dela Cruz Carlo Dominic De Guzman John Philip Tuazon Joash Christian Liwanag
Date Submitted	October 12 2018
Period Covering	October 9 2018 TO October 12 2018
Over-all Project Status (%)	5%

Status	Milestone Description
	CANVASSING THE PRICE OF MATERIALS NEEDED
	IN OUR DEVICE AND GATHER SOME INFORMATION
	FOR THE MATERIALS THAT WE WILL BE USING IN
	OUR PROTOTYPE

Highlight Progress

	Arduino Uno R3	Item	1	Item		P540.00 80640 5%
	RFID Card	Item	1	Item		
	RFID Reader	Item	1	Item		
		Item	1	Item		P741.00 80640 5%
		Item	1	Item		P20.00



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RESEARCHER'S WEEKLY PROGRESS REPORTS

AMA COMPUTER COLLEGE-MALOLOS

**ECE 201:DESIGN PROJECT 2
WEEKLY PROGRESS REPORT**

PROJECT TITLE:	Automated Library Management System using Barcode and RFID
-----------------------	---

PROPONENTS:	Melvin Dela Cruz Carlo Dominic De Guzman John Philip Tuazon Joash Christian Liwanag
Date Submitted	October 19, 2018
Period Covering	October 15, 2018 TO October 19, 2018
Over-all Project Status (%)	7%

Status	Milestone Description
	Installed Visual Studio and MySQL
	Tested remotng into a different PC running a MySQL server

Highlight Progress

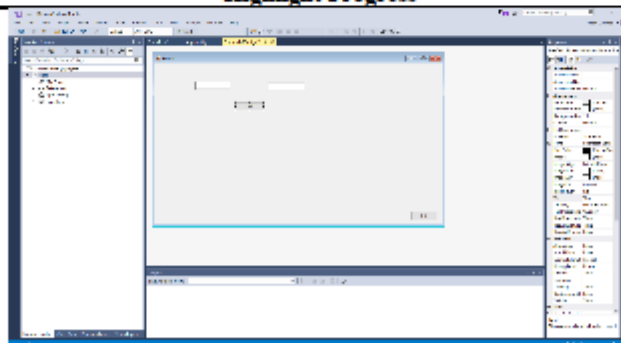


Figure 1. Visual Studio 2017 with a Test Project Open.



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**RESEARCHER'S WEEKLY PROGRESS REPORTS
AMA COMPUTER COLLEGE-MALOLOS**

**ECE 201:DESIGN PROJECT 2
WEEKLY PROGRESS REPORT**

PROJECT TITLE:	Automated Library Management System using Barcode and RFID		
PROPENENTS:	Melvin Dela Cruz Carlo Dominic De Guzman John Philip Tuazon Joash Christian Liwanag		
Date Submitted	October 19, 2018		
Period Covering	October 22, 2018	TO	October 26, 2018
Over-all Project Status (%)	10%		

Status	Milestone Description
	Purchasing of Items
	Connecting the MySQL Database to Visual Studio

Highlight Progress

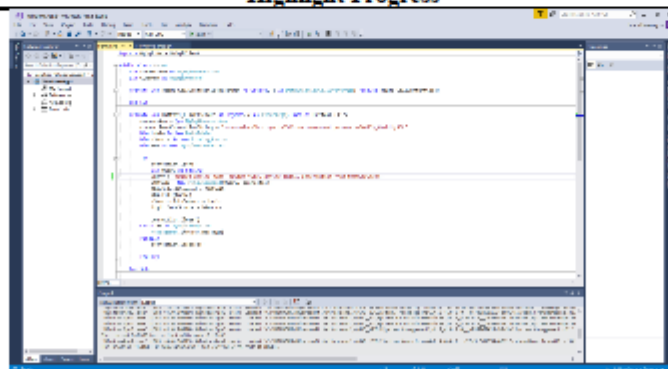


Figure 1. Visual Basic code to connect to MySQL.



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AMA COMPUTER COLLEGE-MALOLOS

**ECE 201:DESIGN PROJECT 2
WEEKLY PROGRESS REPORT**

PROJECT TITLE:	Automated Library Management System using Barcode and RFID
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PROPONENTS:	Melvin Dela Cruz Carlo Dominic De Guzman John Philip Tuazon Joash Christian Liwanag
Date Submitted	November 9, 2018
Period Covering	November 5, 2018 TO November 9, 2018
Over-all Project Status (%)	25%

Status	Milestone Description
Working	RFID Test
In Progress	Login Interface and Main Interface
In Progress	Database
	Highlight Progress

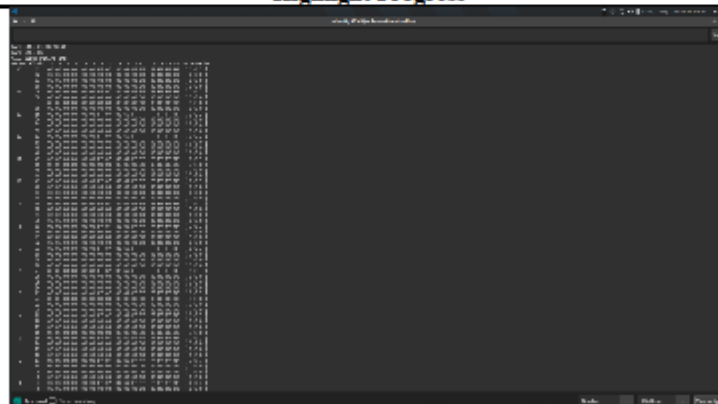


Figure 1. Serial monitor output of the RFID.



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RESEARCHER'S WEEKLY PROGRESS REPORTS

AMA COMPUTER COLLEGE-MALOLOS

**ECE 201:DESIGN PROJECT 2
WEEKLY PROGRESS REPORT**

PROJECT TITLE:	Automated Library Management System using Barcode and RFID		
PROPONENTS:	Melvin Dela Cruz Carlo Dominic De Guzman John Philip Tuazon Joash Christian Liwanag		
Date Submitted	November 23, 2018		
Period Covering	November 12, 2018	TO	November 23, 2018
Over-all Project Status (%)	50%		

Status	Milestone Description
Working	RFID and Barcode test
In Progress	Reading of the RFID UID on Visual Studio
Purchased	USB Barcode Scanner
Highlight Progress	

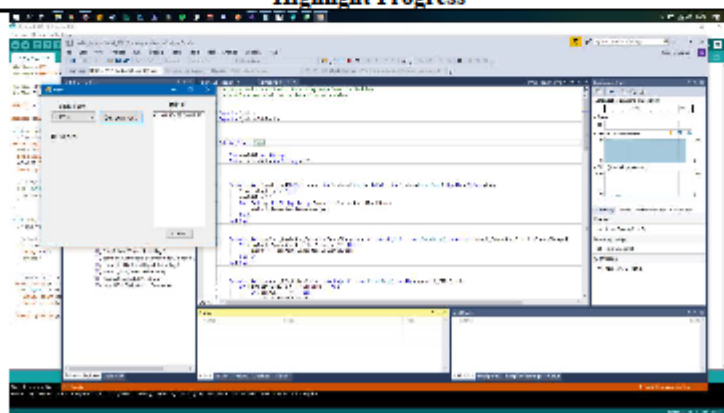


Figure 1. Reading of Serial output on a test Visual Basic program through COM Port 3.

C. DOCUMENTATION AND PHOTOS





The researcher's initial testing of the RFID and the Barcode Scanner



The researchers during the final system testing.



The researchers with their panelist (middle) after their final defense.

D. DEFINITIONS OF TERMS

Library - a building or room containing collections of books, periodicals, and sometimes films and recorded music for people to read, borrow, or refer to.

Borrower – also called the user, the person borrowing a book from the library either for reading in the library or to checkout.

Librarian - a person, typically with a degree in library science, who administers or assists in a library.

Attendance - handles the logging of users going in and/or out of the library.

RFID – Radio Frequency Identification, denoting technologies that use radio waves to identify people or objects carrying encoded microchips.

Issuing – the process of borrowing/checking-out a book.

Returning – the process of returning a checked-out book.

Arduino - an open-source hardware and software company, project and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control both physically and digitally.



Microsoft Access - a database management system from Microsoft that combines the relational Microsoft Jet Database Engine with a graphical user interface and software-development tools.

Visual Basic - a third-generation event-driven programming language from Microsoft for its Component Object Model programming model first released in 1991 and declared legacy during 2008. Microsoft intended Visual Basic to be relatively easy to learn and use.

Visual Basic .NET - a multi-paradigm, object-oriented programming language, implemented on the .NET Framework. Microsoft launched VB.NET in 2002 as the successor to its original Visual Basic language.

Microsoft Visual Studio - an integrated development environment from Microsoft. It is used to develop computer programs, as well as websites, web apps, web services and mobile apps.

Barcode - a visual, machine-readable representation of data; the data usually describes something about the object that carries the barcode. Traditional barcodes systematically represent data by varying the widths and spacings of parallel lines and may be referred to as linear or one-dimensional (1D). Later, two-dimensional (2D) variants were developed, using rectangles, dots, hexagons and other geometric patterns, called *matrix codes* or *2D barcodes*, although they do not use bars as such.

E. REFERENCES

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Journal of Interlibrary Loan, Document Delivery and Electronic Reserve, 18(4), 439-447, <http://www.tandfonline.com/doi/pdf/10.1080/10723030.802181778>



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<https://www.pcworld.com/article/2011352/texas-school-uses-rfid-badges-to-track-student-locations.html>

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(RFID To Go)<http://www.dost.gov.ph/knowledge-resources/news/33-2013-news/186-rfid-technology-to-complement-dost-s-smarter-philippines-program.html>



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COMPUTER COLLEGE OF MALOLOS COLLEGE OF ENGINEERING

F. Curriculum Vitae

CARLO DOMINIC MARQUEZ-DE GUZMAN

129 Sitio Silangan, Pinagkuartelan, Pandi, Bulacan
0956-235-2487
carlo_dg1@yahoo.com



OBJECTIVE

To gain more knowledge and skills in the field of Electronics and Communication Engineering, that will strengthen my education and professional skills and to significantly contribute to the company's organizational objectives.

EDUCATION

- Bachelor of Science Electronics and Communication Engineering
AMA Computer College Malolos
Sumapang Matanda, Malolos City
A.Y. 2015-present
- Bachelor of Science in General Engineering
Bulacan State University-Bustos Campus
Bustos, Bulacan
A.Y. 2014-2015
- Balagtas National Agricultural High School
Pulong Gubat, Balagtas, Bulacan
A.Y. 2010-2014
- Gentle Shepherd Montessori
Siling Matanda, Pandi, Bulacan
A.Y. 2004-2010

ACHIEVEMENTS/ RESPONSIBILITIES

- President and Head BOD of IECEP-AMACC Malolos SC local organization
A.Y. 2018-2019
- Vice President of IECEP-AMACC Malolos SC local organization
A.Y. 2017-2018
- Second trimester Dean's lister
A.Y. 2017-2018
- First trimester Dean's lister
A.Y. 2017-2018
- Second trimester Dean's lister
A.Y. 2016-2017
- Computer hardware Servicing NCII Passer
March 2014

SKILLS

- Proficient of creating Local area network.
- Proficient of trouble shooting computer hardware, and installing operating system.
- Capable of trouble shooting electronic device.
- Proficient of using Microsoft Office applications
- Capable of using software applications such as Multisim, Picosoft, and Fluidsim.

REFERENCES

- Engr. Paul Ryan A. Santiago
Instructor, AMA Computer
College Malolos
0905-131-3357
- Engr. Atilano C. Mendoza
Businessman
0936-924-1914

SEMINARS ATTENDED

- IECEP NATIONAL STUDENT
SUMMIT 2018
University of the Philippines,
Quezon city
November 10, 2018
- Restricted Radiotelephone
Operator's (RLM) Certificate
for Land Mobile Stations,
Digital Mobile Radio, Amateur
Radio, First Class
Radiotelephone Operator)
Polytechnic University of the
Philippines-Mabini Campus,
Santa Mesa, Manila
October 22, 2018
- Solar Photovoltaic System
Workshop
Malolos City
Dec 2017
- IECEP-BulSU SC Excelerate
Seminar 2017
Bulacan State University
November 18, 2017
- Basic Electro-Pneumatics
AMACC-MALOLOS
October 10, 2017
- Fluke Networks CCTT
Associate Program
Bulacan State University
March 26, 2017

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

CARLO DOMINIC M. DE GUZMAN
Applicant



COMPUTER COLLEGE OF MALOLOS COLLEGE OF ENGINEERING



JOHN PHILIP NIETO TUAZON

1164 Maligaya St. Pulong Buhangin, Sta. Maria, Bulacan

+63 977 334 7325 | p.tuazon001@gmail.com

OBJECTIVE

To attain an internship program which will help me to have an experience and to cope in the fields of Electronics and Communication Engineering so that I can learn new things and enhance my skills as I gain experience.

TECHNICAL SKILLS

- Hardware troubleshooting
- Network troubleshooting
- Proficient of using Microsoft Office applications

PERSONAL SKILLS

- Good written and verbal communication skills
- Collaborative and able to work in teams or groups
- Motivated and willing to learn new things

EDUCATION

- Bachelor of Science Electronics and Communication Engineering
AMA Computer College Malolos
Sumapang Matanda, Malolos City
A.Y. 2015-present
- Immaculate Heart of Mary Integrated School
Pulong Buhangin, Sta. Maria, Bulacan
A.Y. 2011-2015
- Immaculate Heart of Mary Integrated School
Pulong Buhangin, Sta. Maria, Bulacan
.....

REFERENCE

- Engr. Paul Ryan A. Santiago
Instructor, AMA Computer College Malolos
0905-131-3357
- Engr. Louieanne Angeles
Software Engineering Team Lead, Accenture Inc.
0916-709-9593

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

John Philip N. Tuazon
Applicant



COMPUTER COLLEGE OF MALOLOS COLLEGE OF ENGINEERING

Joash Christian Bautista Liwanag



OBJECTIVE

To attain an internship program to gain experience and further learn skills in the field of Electronics Engineering.

EXPERIENCE

AUDIO TECHNICIAN

RANM, Malolos City / 2013 - Present

- FOH Technician for a Church every Sunday since 2013.
- Troubleshoot and maintain Audio Equipment.

EDUCATION

BACHELOR OF SCIENCE IN ELECTRONICS AND COMMUNICATIONS ENGINEERING

AMA Computer College Malolos, Malolos City

A.Y. 2016 – present

BACHELOR OF SCIENCE IN ELECTRONICS AND COMMUNICATIONS ENGINEERING

Ateneo de Manila University, Quezon City

A.Y. 2014 – 2016

HIGH SCHOOL DEGREE

Darwin International School, Malolos City

A.Y. 2010 - 2014

CONTACT

Email:
joashliwanag@gmail.com

Phone:
+639560560780

Address:
Blk 1, Lot 8, Ph 2A, Provence
Subd. Longos, Malolos City

Github Profile:
github.com/ajigsawnhalo

SOFT SKILLS

Adaptability

Collaboration

Strong Work Ethic

Problem Solving /
Troubleshooting

HARD SKILLS

Proficient in C++

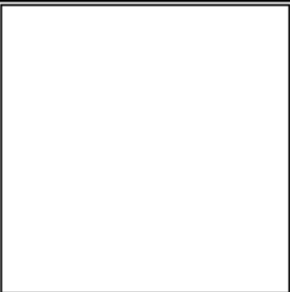
Proficient in Linux

Network Troubleshooting

Microsoft Office



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	QUALIFICATIONS PROFILE Proficient and thoughtful professional prepared to excel in the field of electronics engineering and significantly contribute to the company's organizational objectives. Key Strengths • Administration and Communication: Always prioritize and organize workloads to meet deadlines. Establishes and maintains strong working relationships with the management, staffs and external groups. • Technical Proficiencies: Capable of using Microsoft Office applications and other software applications such as MultiSIM and MatLab.
	CHARACTER REFERENCES • Mr. Raevenor Gonzalez Instructor Bulacan State University – Bustos Campus Bustos, Bulacan (0925)-830-3884 • Mr. Joseph Emmanuel S. Guevara Dean

MELVIN C. DELA CRUZ Pandí, Bulacan 0915-867-3291 melvindelacruz110994n@gmail.com									
PERSONAL INFORMATION Age: 24 Date of birth: September 11 1994 Place of birth: Bulacan Gender: Male Height: (5'9") Weight: (57kg) Civil status: Single Religion: Catholic									
EDUCATION <table><tr><td>Tertiary</td><td>Bachelor of Science in Electronics Engineering AMA Computer College Malolos City of Malolos, Bulacan (S.Y. 2015 - Present)</td></tr><tr><td></td><td>Associate in Computer Technology STI Balagtas, Bulacan City of Malolos, Bulacan (S.Y. 2011 - 2013)</td></tr><tr><td>Secondary</td><td>Holy Angels College Pandí Bulacan (S.Y. 2007- 2011)</td></tr><tr><td>Primary</td><td>M.C Bernardo Memorial Central School Pandí, Bulacan (S.Y. 2000-2007)</td></tr></table>		Tertiary	Bachelor of Science in Electronics Engineering AMA Computer College Malolos City of Malolos, Bulacan (S.Y. 2015 - Present)		Associate in Computer Technology STI Balagtas, Bulacan City of Malolos, Bulacan (S.Y. 2011 - 2013)	Secondary	Holy Angels College Pandí Bulacan (S.Y. 2007- 2011)	Primary	M.C Bernardo Memorial Central School Pandí, Bulacan (S.Y. 2000-2007)
Tertiary	Bachelor of Science in Electronics Engineering AMA Computer College Malolos City of Malolos, Bulacan (S.Y. 2015 - Present)								
	Associate in Computer Technology STI Balagtas, Bulacan City of Malolos, Bulacan (S.Y. 2011 - 2013)								
Secondary	Holy Angels College Pandí Bulacan (S.Y. 2007- 2011)								
Primary	M.C Bernardo Memorial Central School Pandí, Bulacan (S.Y. 2000-2007)								

<i>I hereby certify that the above information is true and correct to the best of my knowledge and belief.</i> Melvin C. Dela Cruz



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G. Evaluation Sheets